Adaptive Genius as a Structural Viability Regime

A Dynamical-Systems Formalization of Cognitive Potential Realization

Boris Kriger[1,2]

[1] Information Physics Institute, Gosport, Hampshire, United Kingdom

[2] Institute of Integrative and Interdisciplinary Research, Toronto, Canada

ORCID: 0009-0001-0034-2903

February 2026

Abstract

Geniality is widely discussed in philosophy, psychology, and neuroscience, yet it lacks a unified formal framework connecting its many contributing factors—genetic endowment, environment, deliberate practice, self-belief, and technological augmentation—into a single mathematical structure. This paper proposes such a framework by modelling the realization of cognitive potential as a dynamical system with multiple interacting state variables, threshold effects, and feedback loops. We define the *genius regime* as a structurally stable region in the system's phase space where plasticity, self-belief, environmental coupling, practice intensity, and motivational drive jointly satisfy viability conditions. We derive necessary conditions under which the system enters this regime, characterize two competing attractors (a *stagnation basin* and a *growth trajectory*), and show that the transition between them exhibits properties of a bifurcation governed by a self-belief metavariable. The model further incorporates artificial intelligence as a cognitive extension module with a bandwidth parameter, formalizing the "bottleneck" between human thought and machine processing. Historical convergence of discoveries is modelled via a complementary percolation framework on knowledge graphs. The formalism is intended as a *conceptual model*—a coherent mathematical language that generates structural hypotheses and identifies intervention points, rather than a calibrated predictive theory. All mathematical content is presented with extensive conceptual commentary to remain accessible to readers from psychology, sociology, and biology. The paper situates itself within the Unified Structural Theory of Complex Systems as a contribution to the cognitive architecture layer, filling a gap between the theory's dynamical core and its applications to social and technological systems.

Keywords: cognitive potential, dynamical systems, viability theory, neuroplasticity, genius, self-organization, artificial intelligence, creativity, phase transitions, complex systems

Contents

1 Introduction

What makes someone a genius? The question has occupied philosophers, psychologists, and neuroscientists for centuries, yet the answers remain fragmented. Geneticists point to heritable traits [11], psychologists emphasize deliberate practice [3] and growth mindset [26], neuroscientists study synaptic density and dopaminergic pathways, and sociologists highlight the role of culture and historical timing. Each perspective captures a piece of the phenomenon, but no existing framework integrates these pieces into a single, formally coherent structure.

This paper proposes such a framework. We treat the realization of cognitive potential as a *dynamical system*—a mathematical object that describes how a collection of interrelated quantities evolves over time according to specified rules. Within this system, what is colloquially called "genius" emerges not as a fixed trait but as a *regime*: a particular mode of operation in which several variables—genetic endowment, environmental quality, practice intensity, self-belief, internal motivation, and technological augmentation—interact in a mutually reinforcing manner.

The key insight, drawn from the philosophical and neuroscientific analysis in [68], is that the factors contributing to genius are not additive. A person with extraordinary genetic endowment but no access to education does not achieve half the output of a genius with both; rather, the output may be negligible. Conversely, moderate endowment combined with high discipline, a supportive environment, and modern technological tools can produce results that rival or exceed those of the classically "gifted." This pattern—where factors multiply rather than add, and where thresholds separate qualitatively different outcomes—is precisely what dynamical systems theory is designed to capture.

1.1 Aims and scope

The specific goals of this paper are:

(i) To define the state variables and parameters of a dynamical system whose trajectories represent the temporal evolution of an individual's realized cognitive potential.

(ii) To characterize the *genius regime* as a structurally stable attractor in this system's phase space, and to derive the conditions under which the system enters and remains in this regime.

(iii) To formalize the role of self-belief as a *metavariable*—a quantity that governs the dynamics of other variables—and to show that it controls a bifurcation between stagnation and growth.

(iv) To model historical convergence of discoveries as a threshold phenomenon on knowledge networks.

(v) To incorporate artificial intelligence as a cognitive extension and to identify the bandwidth of the human–AI interface as the critical parameter determining the effectiveness of this augmentation.

(vi) To present all mathematical content with thorough conceptual explanations so that the paper remains accessible to researchers in psychology, sociology, and biology who may not have backgrounds in dynamical systems.

1.2 Structure of the paper

Section 2 provides a comprehensive review of the relevant literature across six contributing disciplines. Section 3 introduces the core dynamical system and its state variables. Section 4 defines the viability kernel—the set of conditions under which the genius regime is sustained—and derives the main existence result. Section 5 analyzes the role of self-belief as a bifurcation parameter. Section 6 connects the formalism to neuroscientific mechanisms. Section 7 models historical convergence. Section 8 formalizes cognitive augmentation through AI. Section 9 treats creativity as a balance between divergent generation and convergent selection. Section 13 situates the paper within the broader Unified Structural Theory. Section 15 concludes.

2 Literature Review

The phenomenon of genius has been studied from multiple disciplinary perspectives, each illuminating distinct aspects of exceptional cognitive performance. This section reviews six streams of research that collectively provide the empirical and theoretical foundation for the formal model developed in subsequent sections. Throughout, we identify the specific insights that each tradition contributes to the model's variables, parameters, and structural assumptions.

2.1 Expertise and deliberate practice

The modern scientific study of expertise begins with the work of de Groot [1], who showed that chess masters differ from amateurs not in raw processing speed but in the richness of their pattern libraries—an early indication that expertise is a product of structured experience rather than fixed endowment. This finding was extended by Chase and Simon [2], who estimated that achieving master-level chess performance requires the accumulation of approximately 50,000 meaningful patterns, built through years of study and play.

Ericsson, Krampe, and Tesch-Römer [3] crystallized this tradition into the theory of *deliberate practice*, arguing that the primary predictor of expertise across domains—music, chess, sports, mathematics—is the cumulative amount of focused, effortful practice directed at improving specific aspects of performance. Their "10,000-hour rule," though subsequently nuanced [4, 5], established that practice intensity is a necessary (though not sufficient) condition for exceptional achievement. Macnamara et al. [4] conducted a meta-analysis of 88 studies and found that deliberate practice accounts for approximately 12% of variance in performance across domains, with larger effects in highly structured domains (games, music) and smaller effects in less structured ones (professional performance, education). Hambrick et al. [5] argued that additional factors—working memory capacity, age of onset, and domain-specific aptitudes—explain substantial additional variance.

In our model, this stream of research motivates the state variable $P(t)$ (deliberate practice intensity) and the multiplicative coupling between P and the hereditary parameter H: practice matters enormously, but its effectiveness is modulated by the neural substrate on which it operates. The meta-analytic finding that practice explains a variable fraction of performance variance across domains is consistent with the multiplicative (rather than additive) structure of our model, where the marginal effect of practice depends on the levels of all other variables.

2.2 Intelligence, heritability, and the genetics of cognition

Twin and adoption studies have consistently shown that general cognitive ability has substantial heritability, with estimates ranging from 0.50 to 0.80 in adults [6, 7]. Bouchard and McGue [7] reported a correlation of 0.72 between monozygotic twins reared apart, establishing that genetic factors contribute significantly to cognitive ability. Genome-wide association studies (GWAS) have identified hundreds of genetic variants associated with educational attainment and cognitive performance, though each individual variant contributes a tiny fraction of the total variance [8, 9]. Savage et al. [9] identified 205 genomic loci in a sample of 269,867 individuals, but found that all identified variants together explained less than 5% of phenotypic variance, revealing the highly polygenic architecture of intelligence.

The sociologist Dean Keith Simonton [10, 11] provided a historiometric analysis of genius, treating it as an emergent property of multiple factors—ability, productivity, and zeitgeist—operating multiplicatively. His "chance-configuration theory" [10] models creative genius as the result of blind variation and selective retention operating on mental elements, a Darwinian process that depends on the size and diversity of the combinatorial space available to the individual. Eysenck [12] similarly argued that genius requires the convergence of high intelligence, personality traits (especially psychoticism as a driver of loose associations), and social factors, with the conjunction of all three being exceedingly rare.

These findings motivate our hereditary parameter H as a composite, polygenic quantity that sets an upper bound on the *rate* of neural restructuring, and they support the multiplicative structure of the governing equations: the rarity of genius arises not from the rarity of any single trait but from the rarity of their simultaneous conjunction at sufficiently high levels.

2.3 Neuroplasticity and the biological basis of learning

The discovery that the adult brain retains significant structural and functional plasticity [15, 16] overturned the earlier view that neural circuits are fixed after critical periods. Pascual-Leone et al. [16] reviewed evidence from neuroimaging, transcranial magnetic stimulation, and lesion studies showing that cortical maps reorganize in response to experience throughout the lifespan, with the rate of reorganization declining with age but never reaching zero. Draganski et al. [17] demonstrated grey matter increases in juggling trainees after only three months of practice, providing direct evidence that macroscopic brain structure changes in response to skill acquisition.

At the synaptic level, Hebb's postulate [13]—that neurons that fire together strengthen their connections—was formalized by Bliss and Lømo [14] through the discovery of long-term potentiation (LTP). Subsequent work identified prediction-error signalling in the dopaminergic system as a key mechanism for learning: Schultz, Dayan, and Montague [18] showed that midbrain dopamine neurons encode the discrepancy between predicted and actual rewards, providing a biological implementation of the temporal-difference learning algorithm from reinforcement learning theory [19].

The brain's default mode network (DMN), identified by Raichle et al. [20] and further characterized by Buckner, Andrews-Hanna, and Schacter [21], has been linked to spontaneous thought, mind-wandering, and the generation of novel associations. Beaty et al. [22] showed that highly creative individuals exhibit stronger functional connectivity between the DMN and the executive control network (ECN), suggesting that creativity depends on the coordinated interaction of spontaneous idea generation and deliberate evaluation.

These findings ground our plasticity coefficient $\alpha(t)$ (equation 6), the prediction-error learning rule (Definition 6.1), and the divergent–convergent creativity model (Section 9).

2.4 Self-efficacy, growth mindset, and the psychology of belief

Bandura's [24, 25] theory of self-efficacy established that an individual's belief in their ability to execute the actions required for specific outcomes is a powerful predictor of effort, persistence, and achievement. Meta-analyses have confirmed strong correlations between self-efficacy and academic performance [27], with effect sizes comparable to those of cognitive ability itself. Stajkovic and Luthans [27], in a meta-analysis of 114 studies comprising 21,616 subjects, found a weighted average correlation of $r = 0.38$ between self-efficacy and work-related performance.

Dweck's [26] distinction between fixed and growth mindsets—beliefs about whether intelligence is innate and immutable or developable through effort—has generated extensive research and debate. Yeager et al. [28] conducted a nationally representative randomized trial in the United States and found that a brief growth-mindset intervention improved grades among lower-achieving students, particularly in schools with supportive peer norms. However, Sisk et al. [29] conducted meta-analyses finding weak overall relationships between mindset and academic achievement ($r = 0.10$), though effects were stronger in at-risk populations.

The impostor syndrome literature [30, 31] documents the converse phenomenon: individuals with objectively high ability who persistently doubt their competence, experiencing anxiety and underperformance as a result. Clance and Imes [31] first described the "impostor phenomenon" among high-achieving women, and Bravata et al. [30] estimated prevalence at 9–82% depending on the population and screening instrument.

In our model, these findings motivate the self-belief variable $S(t)$ and its dynamics (equation 4). The bifurcation at $S = S_c$ (Proposition 5.1) formalizes the observation that self-belief creates self-reinforcing cycles: those who believe they can improve actually do (through increased practice and persistence), while those who believe ability is fixed disengage in the face of difficulty, confirming their own pessimistic expectations. The impostor syndrome corresponds to states near the boundary of the viability kernel where S is close to S_c despite high values of other variables.

2.5 Intrinsic motivation, flow, and self-determination theory

Deci and Ryan's [32, 33] self-determination theory identifies three basic psychological needs—autonomy, competence, and relatedness—whose satisfaction fosters intrinsic motivation, while their frustration promotes amotivation or extrinsically regulated behaviour. Intrinsic motivation—engagement in activity for its inherent interest and enjoyment—has been linked to deeper learning, greater creativity, and enhanced persistence [36].

Csikszentmihalyi's [34, 35] concept of "flow" describes a state of complete absorption in a challenging activity matched to one's skill level, characterized by loss of self-consciousness, distorted time perception, and intrinsic reward. Flow occurs at the boundary between boredom (skill exceeds challenge) and anxiety (challenge exceeds skill), and is associated with enhanced performance across domains. Amabile's [36] componential theory of creativity identifies intrinsic motivation as one of three essential components (alongside domain-relevant skills and creativity-relevant processes), arguing that extrinsic motivators—especially those perceived as controlling—can undermine creative output.

These findings motivate the intrinsic motivation variable $M(t)$ and its dynamics (equation 5), particularly the suppressive effect of external coercion $R(t)$ on motivation. The flow state corresponds to a region of the state space where P, M, S, and G are simultaneously high and mutually reinforcing.

2.6 Creativity research: divergent thinking, remote associations, and network models

Guilford [37, 38] introduced the concept of divergent thinking—the ability to generate multiple solutions to open-ended problems—as a core component of creativity, distinguishing it from convergent thinking (finding the single correct answer). The Torrance Tests of Creative Thinking [39], based on Guilford's framework, remain among the most widely used assessments of creative potential, measuring fluency (number of ideas), flexibility (number of categories), originality (statistical rarity of ideas), and elaboration (level of detail).

Mednick [40] proposed the associative theory of creativity, arguing that creative individuals have flatter associative hierarchies—they can access remote associations more readily than non-creative individuals. The Remote Associates Test (RAT) was developed to operationalize this construct. More recently, Kenett et al. [41] used network science to model semantic memory as a graph, finding that creative individuals have more flexible, interconnected, and less modular semantic networks than less creative individuals.

Campbell [42] and Simonton [10] independently developed "blind variation and selective retention" (BVSR) models of creativity, treating ideation as a quasi-random generative process followed by evaluative selection. This two-stage model—generate, then select—is widely accepted in creativity research [43] and provides the conceptual basis for our divergent–convergent formalization (Section 9).

Martindale [44] proposed a neurophysiological theory of creativity based on cortical arousal, arguing that low arousal (defocused attention) facilitates the generation of remote associations, while high arousal (focused attention) facilitates their evaluation. This arousal-based account anticipates the neuroimaging findings of Beaty et al. [22] on DMN–ECN coupling and provides physiological grounding for the D/F_c balance in our creativity model.

2.7 Environmental and sociocultural factors

Bronfenbrenner's [45] ecological systems theory emphasizes that development occurs within nested environmental contexts—microsystem, mesosystem, exosystem, macrosystem—each exerting distinct influences on the individual. Applied to genius, this framework highlights that exceptional achievement is not solely an individual property but emerges from the interaction between individual capacities and environmental affordances at multiple scales.

Heckman [46] and Cunha and Heckman [47] provided econometric evidence that early childhood investments in cognitive and non-cognitive skills exhibit strong complementarities: the return on investment in later skills depends critically on the level of earlier skills, and vice versa. This "skill begets skill" pattern—a self-reinforcing dynamic in which early advantages compound over time—is formally analogous to the multiplicative coupling in our model and to the Matthew effect in the environmental equation (3).

Merton's [48] "Matthew effect in science"—the observation that eminent scientists receive disproportionate credit for their contributions, while unknown scientists receive disproportionately little—demonstrates how environmental feedback amplifies initial differences. Zuckerman [50] documented this effect empirically among Nobel laureates, showing that early career advantages (prestigious mentors, elite institutions) compound through preferential access to resources, visibility, and collaborative opportunities.

The concept of *multiple discoveries*—the independent, near-simultaneous arrival at the same idea by different researchers—was systematically studied by Ogburn and Thomas [51], who documented 148 instances, and by Merton [49], who argued that such "multiples" are the rule rather than the exception in science. Lamb and Easton [52] provided additional historical analysis. These findings motivate our percolation model of convergence (Section 7).

2.8 Human–AI interaction and cognitive augmentation

The concept of human–computer symbiosis was articulated by Licklider [54], who envisioned a partnership in which human and machine each contribute their distinctive strengths: humans providing goals, context, and judgment; machines providing speed, precision, and data processing. This vision has been revived in the era of large language models and neural interfaces.

Engelbart [55] proposed a framework for "augmenting human intellect" through technological tools, emphasizing that augmentation is not merely about automation but about expanding the repertoire of cognitive strategies available to the human. More recently, the "intelligence augmentation" (IA) paradigm [57] has been contrasted with the "artificial intelligence" (AI) paradigm, with IA focusing on systems that enhance human capabilities rather than replacing them.

The bandwidth constraint on human–computer interaction was quantified by Card, Moran, and Newell [56], who estimated the human information-processing rate at approximately 7–10 bits per second for deliberate cognitive tasks. This severe bottleneck motivates our bandwidth parameter $B(t)$ and the integration coefficient $\beta(t)$ (Section 8). Neuralink and related brain–computer interface (BCI) programmes aim to increase this bandwidth by several orders of magnitude, potentially transforming the dynamics of human–AI collaboration [58].

The extended mind thesis of Clark and Chalmers [59] provides philosophical grounding for treating AI as a genuine cognitive extension rather than a mere tool: if the relevant functional relationships are in place, external devices that participate in cognitive processes should be considered part of the cognitive system itself. This view supports our treatment of the human–AI system as a single dynamical entity with total capacity K_{total} (Definition 8.1).

2.9 Dynamical systems and viability theory in cognitive science

The application of dynamical systems theory to cognition has a rich history. Thelen and Smith [60] pioneered the dynamical systems approach to development, showing that motor and cognitive milestones emerge from the self-organization of multiple interacting subsystems rather than from the unfolding of a genetic program. Port and van Gelder [61] argued that cognitive processes are best understood as trajectories in state spaces rather than as symbol manipulations, and Kelso [62] demonstrated coordination dynamics—bifurcations, phase transitions, multistability—in sensorimotor behaviour.

Viability theory, developed by Aubin [63, 64] in the context of differential inclusions and control theory, provides the mathematical framework for determining whether a dynamical system can remain within a prescribed constraint set. The viability kernel—the largest set of initial conditions from which at least one trajectory remains viable—has been applied to ecological management [65], fisheries [66], and economic sustainability [67], but has not previously been applied to the realization of cognitive potential.

Our model extends the viability-theoretic framework to the cognitive domain, treating genius not as an optimization problem ("maximize output") but as a viability problem ("maintain all essential variables within their admissible ranges simultaneously"). This reframing is essential because the multiplicative coupling between variables means that maximizing one at the expense of others (e.g., practice to the point of burnout, destroying motivation) is counterproductive.

2.10 Summary: the gap this paper fills

Despite the richness of these individual research traditions, no existing framework combines them into a single formal structure. Expertise research focuses on practice but underspecifies the role of self-belief and environment. Intelligence genetics quantifies heritability but says little about how potential is converted into achievement. Creativity research models ideation processes but does not connect them to the broader trajectory of intellectual development. Motivation theory explains why people engage or disengage but does not formalize the feedback loops with environmental and cognitive variables.

The present paper fills this gap by providing a dynamical-systems model that integrates all six streams—expertise, genetics, neuroplasticity, self-belief, motivation, and environment—into a single set of coupled equations whose qualitative behaviour (two attractors, a bifurcation, multiplicative interactions, viability conditions) captures the empirically observed patterns in a formally rigorous and testable manner.

3 The Dynamical Framework

For readers unfamiliar with the mathematical formalism, a *dynamical system* is simply a precise way of describing how things change over time. Imagine tracking multiple quantities simultaneously—say, a person's skill level, their motivation, and the quality of their environment. A dynamical system specifies, for any given combination of these quantities at a given moment, *how fast and in what direction* each quantity is changing. The mathematical tool for this is a system of *differential equations*, which are rules that relate the rate of change of each quantity to the current values of all the quantities. This approach has been applied to cognitive and developmental science by Thelen and Smith [60] and to coordination dynamics by Kelso [62], but has not previously been applied to the realization of cognitive potential as a viability problem in the sense of Aubin [63].

The complete collection of all possible combinations of the quantities is called the *state space*: it is an abstract landscape in which the system's condition at any moment corresponds to a single point. As time passes, this point traces out a curve called a *trajectory*. Special points or regions where trajectories tend to converge are called *attractors*—they represent the long-run states toward which the system naturally evolves.

With this background in mind, we now define the specific system that models the realization of cognitive potential.

3.1 State variables

We model the cognitive system of an individual at time t by a vector of five state variables, each taking non-negative real values:

Definition 3.1 (State variables of the cognitive potential system)**.** Let $t \geq 0$ denote time. The **cognitive potential system** is described by the state vector

$$\mathbf{x}(t) = \big(G(t),\ P(t),\ E(t),\ S(t),\ M(t)\big) \in \mathbb{R}^5_{\geq 0},$$

where each component has the following interpretation:

- $G(t)$: **Realized intellectual output** — the cumulative, quality-weighted level of the individual's cognitive achievement at time t. This is not a fixed "IQ" but a dynamic measure that grows (or stagnates) depending on all other variables.

- $P(t)$: **Deliberate practice intensity** — the rate at which the individual engages in effortful, structured cognitive work at time t. This corresponds to the concept of "deliberate practice" in expertise research [3]: not mere repetition, but focused effort directed at improving performance. The distinction between deliberate practice and mere experience is critical; as Macnamara et al. [4] showed in their meta-analysis, the former predicts performance far more strongly than the latter.

- $E(t)$: **Environmental quality** — a composite measure of the external conditions available to the individual: access to education, quality of mentorship, cultural richness, availability of resources, and freedom from oppressive constraints. This variable captures the nested environmental systems described in Bronfenbrenner's [45] ecological model and the skill-complementarity effects documented by Cunha and Heckman [47].

- $S(t)$: **Self-belief** — the individual's subjective assessment of their own capacity for intellectual growth. This corresponds to the "growth mindset" construct [26], to Bandura's [25] self-efficacy theory, and to the social cognitive theory's emphasis on perceived capability as a predictor of effort and persistence [27].

- $M(t)$: **Intrinsic motivation** — the intensity of the individual's internally generated drive to explore, create, and understand, independent of external rewards. This corresponds to the concept of intrinsic motivation in Deci and Ryan's [32, 33] self-determination theory, to the "flow" state described by Csikszentmihalyi [34, 35], and to Amabile's [36] identification of intrinsic motivation as a core component of creative performance.

Remark 3.2. The variable $G(t)$ should not be confused with a measure of "innate talent." Rather, it represents the *realized* intellectual output at time t—the visible, accumulated result of the interaction between all factors. A person with high innate potential but no practice, no environment, and no motivation will have a low value of G.

In addition to the state variables, we introduce a *parameter* that represents genetic and constitutional endowment:

Definition 3.3 (Hereditary parameter)**.** The **hereditary parameter** $H \in [0, 1]$ is a time-invariant quantity representing the individual's genetically and constitutionally determined upper bound on the rate at which neural structures can be modified through experience. Informally, H captures "raw neural hardware quality"—speed of signal processing, baseline synaptic density, neurochemical balance—as reflected in the substantial heritability estimates from twin studies [6, 7] and the highly polygenic architecture revealed by GWAS [8, 9]. Crucially, H sets an upper bound on *rate*, not on *ultimate achievement*: a lower H means slower learning, not a lower ceiling, because the ceiling is determined by the interaction of all variables over time.

We also introduce a *historical context parameter*:

Definition 3.4 (Historical context)**.** The **historical context** $C(t) \geq 0$ represents the cumulative knowledge, technology, and cultural infrastructure available to the individual at time t. This is an exogenous variable—it is determined by the state of civilization rather than by the individual's actions.

3.2 The governing equations

We now specify how the state variables evolve over time. The key principle is *multiplicative interaction*: the factors do not simply add up but amplify or suppress one another.

The following informal summary describes the qualitative content of equations (1)–(5) before they are presented formally:

- **Intellectual output grows** when practice is high, the environment is good, self-belief is strong, and motivation is present—but any factor being near zero can suppress growth almost entirely.
- **Practice intensity** increases when self-belief and motivation are high, but decreases due to fatigue and when the environment discourages effort.
- **Environmental quality** evolves on a slower timescale and responds partly to the individual's output (success attracts resources) and partly to exogenous factors.
- **Self-belief** grows when practice yields positive results (a reinforcing feedback loop) but erodes when outcomes fall short of expectations.
- **Intrinsic motivation** is sustained by novelty and progress and suppressed by stagnation and external coercion.

Definition 3.5 (Governing equations)**.** The cognitive potential system evolves according to:

$$\frac{dG}{dt} = H \cdot \alpha(t) \cdot P(t) \cdot \sigma\big(E(t)\big) \cdot S(t) \cdot \phi\big(M(t)\big) - \delta_G\, G(t), \tag{1}$$

$$\frac{dP}{dt} = \beta_P\, S(t)\, M(t) - \gamma_P\, P(t) + \eta_P\, E(t), \tag{2}$$

$$\frac{dE}{dt} = \beta_E\, G(t) + \xi_E(t) - \gamma_E\, E(t), \tag{3}$$

$$\frac{dS}{dt} = \beta_S \left(\frac{dG/dt}{G(t)+\epsilon}\right) S(t)\big(1 - S(t)\big) - \gamma_S\, \psi\big(F(t)\big), \tag{4}$$

$$\frac{dM}{dt} = \beta_M\, \nu\big(G(t)\big) - \gamma_M\, M(t)\, \big(1 + \kappa\, R(t)\big), \tag{5}$$

where the auxiliary functions and parameters are defined as follows.

Let us explain each equation and its components in detail.

Equation (1): Growth of realized output. This is the central equation of the system. The rate of change of realized intellectual output G is the product of several factors:

- H: the hereditary parameter (Definition 3.3), acting as a scaling constant.
- $\alpha(t) \in (0, 1]$: the **neuroplasticity coefficient**, representing the brain's current capacity to form and strengthen neural connections. Intuitively, $\alpha(t)$ captures how adaptable the brain is at time t. It is high in youth and decreases with age, but remains nonzero throughout life—reflecting the neuroscientific evidence that plasticity, while declining, never vanishes entirely [15–17]. We model this as:

$$\alpha(t) = \alpha_0\, e^{-\lambda t} + \alpha_{\min}, \quad \alpha_0, \alpha_{\min}, \lambda > 0, \quad \alpha_{\min} < \alpha_0. \tag{6}$$

  Here $\alpha_0 + \alpha_{\min}$ is the initial plasticity, λ is the decay rate, and $\alpha_{\min}$ is the residual plasticity that persists even in advanced age.

- $P(t)$: deliberate practice intensity, entering multiplicatively. If $P = 0$, then dG/dt reduces to a decay term, meaning that without practice, accumulated output erodes.
- $\sigma(E)$: an **environmental activation function** that translates raw environmental quality into effective support. We model this as a sigmoid:

$$\sigma(E) = \frac{1}{1 + e^{-k_E (E - E^*)}}, \tag{7}$$

  where E^* is the environmental threshold—the level of environmental quality below which support is negligible—and $k_E > 0$ controls the sharpness of the transition. The sigmoid shape captures a crucial empirical observation: very poor environments (war, extreme poverty, total isolation from knowledge) suppress genius almost completely, while very rich environments yield diminishing returns. The transition between these regimes is sharp but not instantaneous.

- $S(t)$: self-belief, also entering multiplicatively. If $S = 0$, then growth halts regardless of all other factors.

- $\phi(M)$: a **motivation activation function**, modelled as $\phi(M) = M/(M + K_M)$ where $K_M > 0$ is a half-saturation constant. This Michaelis–Menten-type function captures the idea that motivation has strongly diminishing returns: the difference between no motivation and moderate motivation is enormous, but the difference between high and very high motivation is small.

- $\delta_G > 0$: a **decay rate** representing the natural erosion of skills and knowledge that are not maintained. Without ongoing practice, realized output diminishes—corresponding to the well-known "use it or lose it" phenomenon in cognitive science.

Equation (2): Dynamics of practice. Practice intensity increases in proportion to the product $S(t) \cdot M(t)$: one needs both the belief that effort will pay off *and* the internal drive to engage in it. The term $-\gamma_P P(t)$ represents fatigue and natural limits on sustained effort. The term $\eta_P E(t)$ captures the facilitating role of the environment: a well-equipped laboratory, a stimulating intellectual community, or access to good tools makes it easier to sustain practice.

Equation (3): Environmental evolution. The environment evolves on a slower timescale. The term $\beta_E G(t)$ represents the *Matthew effect* [48]: success attracts resources, recognition, and better opportunities, a phenomenon documented empirically by Zuckerman [50] among Nobel laureates. The term $\xi_E(t)$ is an exogenous input representing societal and historical changes (e.g., the founding of a university, an economic boom, or a war that disrupts institutions). The decay term $-\gamma_E E(t)$ reflects the fact that environments deteriorate without maintenance.

Equation (4): Dynamics of self-belief. This equation deserves particular attention because self-belief plays a pivotal role in the model. The first term, $\beta_S (dG/dt)/(G + \epsilon) \cdot S(1 - S)$, is a *logistic growth* term modulated by relative progress. In words: self-belief grows when the individual sees that their output is improving (the ratio dG/dt over $G + \epsilon$ measures the *rate of improvement relative to current level*, and $\epsilon > 0$ is a small constant that prevents division by zero). The logistic factor $S(1 - S)$ ensures that S remains bounded between 0 and 1 and that growth is fastest at intermediate values.

The second term, $-\gamma_S \psi(F(t))$, represents the erosive effect of *failure signals* $F(t)$, which we define as negative feedback—rejected proposals, critical evaluations, social discouragement, and comparison with more successful peers. The function ψ is monotonically increasing: more failure signals produce stronger erosion of self-belief.

Equation (5): Dynamics of intrinsic motivation. The function $\nu(G)$ represents **novelty generation**: as G increases, the individual encounters new phenomena, new questions, new connections between ideas, all of which stimulate intrinsic motivation. We model ν as a concave, increasing function—each new insight stimulates motivation, but the stimulation per unit of progress diminishes. The decay term $-\gamma_M M(1 + \kappa R(t))$ captures two sources of motivational erosion: natural fluctuation ($\gamma_M M$) and the suppressive effect of external coercion $R(t)$, where $R(t)$ represents the intensity of externally imposed, non-intrinsically-motivated demands (routine bureaucratic tasks, financial pressures, rigid curricula).

3.3 Summary of the parameter space

Table 1 summarizes all parameters of the model.

Table 1: Parameters of the cognitive potential dynamical system.

Symbol	Name	Interpretation
H	Hereditary parameter	Genetically determined upper bound on learning rate
α_0	Initial plasticity	Neuroplasticity at birth
$\alpha_{\min}$	Residual plasticity	Minimum lifelong neuroplasticity
λ	Plasticity decay rate	Rate of age-related plasticity decline
E^*	Environmental threshold	Level below which environment suppresses growth
k_E	Environmental sharpness	Steepness of environmental activation
K_M	Motivation half-saturation	Motivation level yielding half-maximal effect
δ_G	Output decay rate	Rate of skill erosion without practice
β_P	Practice gain rate	Sensitivity of practice to belief and motivation
γ_P	Fatigue rate	Rate of practice decay due to exhaustion
η_P	Environmental facilitation	Environment's direct effect on practice ease
β_E	Matthew effect strength	Rate at which success attracts resources
γ_E	Environmental decay rate	Rate of environmental deterioration
β_S	Self-belief gain rate	Sensitivity of self-belief to perceived progress
γ_S	Self-belief erosion rate	Sensitivity of self-belief to failure signals
β_M	Motivation gain rate	Sensitivity of motivation to novelty
γ_M	Motivation decay rate	Natural motivational decay rate
κ	Coercion sensitivity	Amplification of decay by external demands

3.4 Justification of functional form choices

The specific functional forms in equations (1)–(5) were chosen to balance mathematical tractability with empirical plausibility. We briefly justify each choice and note alternatives:

Multiplicative coupling in (1). The product structure $H \cdot \alpha \cdot P \cdot \sigma(E) \cdot S \cdot \phi(M)$ is the central modelling decision. It is motivated by the empirical finding that genius factors exhibit strong complementarities [5, 47]: any factor near zero suppresses the entire product. An additive model $(a_1 H + a_2 P + a_3 E + \ldots)$ would predict that high values of one factor can fully compensate for the absence of another, which is inconsistent with the overwhelming empirical evidence that talent without practice, practice without environment, or environment without motivation each yield negligible output.

Sigmoid for environment (7). The sigmoid activation captures a widely observed pattern: very poor environments suppress development almost completely (war zones, extreme poverty), while very rich environments show diminishing returns. The threshold E^* locates the transition. An alternative would be a piecewise-linear function, which is qualitatively equivalent but less smooth.

Michaelis–Menten for motivation. The saturating form $\phi(M) = M/(M + K_M)$ was chosen because it is the simplest bounded function with the property that the marginal effect of motivation is large at low M and small at high M. A sigmoidal alternative would introduce a threshold below which motivation has essentially no effect; this is plausible but adds a parameter without changing the qualitative dynamics in most regimes. Section 12 discusses how alternative choices affect the results.

Logistic self-belief dynamics (4). The term $S(1 - S)$ ensures $S \in [0, 1]$, which is natural for a quantity interpretable as a subjective probability of success. Any bounded growth model (Gompertz, Richards) would preserve this property while altering the speed of self-belief change near the boundaries.

Linear environmental dynamics (3). This is the simplest choice and is acknowledged as a limitation. A more realistic nonlinear extension is proposed in Section 12, equation (27).

Dimensional analysis. We note that the model operates with *dimensionless* variables throughout: H, S, $\sigma(E)$, and $\phi(M)$ are bounded in $[0, 1]$; G, P, E, and M are measured in arbitrary units whose scales are absorbed into the rate parameters. Dimensional consistency of each equation can be verified by noting that both sides have units of $[x]/[\text{time}]$ where $[x]$ is the unit of the respective state variable.

4 The Viability Kernel and the Genius Regime

4.1 Viability: conceptual overview

In everyday terms, a system is *viable* if it can sustain itself over time—if none of its essential components drops to a level where the whole system collapses. A business is viable if its revenue covers its costs; an ecosystem is viable if no critical species goes extinct; a cognitive system is viable if the individual continues to learn, create, and grow.

Mathematically, viability is defined by specifying a region in the state space—called the *viability kernel*—within which all essential variables remain above their critical thresholds. The central question is: given the dynamics of the system, does the trajectory starting from a particular initial condition remain inside the viability kernel, or does it eventually leave?

4.2 Formal definition

Definition 4.1 (Viability kernel for the genius regime)**.** The **viability kernel** $\mathcal{V} \subset \mathbb{R}^5_{\geq 0}$ is the set of states $\mathbf{x} = (G, P, E, S, M)$ satisfying the following conditions simultaneously:

$$\mathcal{V} = \left\{ \mathbf{x} \in \mathbb{R}^5_{\geq 0} \;\middle|\; \begin{array}{l} G \geq G_{\min}, \\ P \geq P_{\min}, \\ E \geq E^*, \\ S \geq S^*, \\ M \geq M_{\min} \end{array} \right\}, \tag{8}$$

where $G_{\min}$, $P_{\min}$, S^*, and $M_{\min}$ are positive thresholds, and E^* is the environmental threshold from (7).

In words: the genius regime is maintained if and only if the individual's output is above a minimum level, they are practising with at least a minimum intensity, the environment meets a quality threshold, self-belief is above a critical level, and intrinsic motivation has not been extinguished.

Definition 4.2 (Genius regime)**.** The **genius regime** is the dynamical state in which the system trajectory $\mathbf{x}(t)$ satisfies $\mathbf{x}(t) \in \mathcal{V}$ for all $t \geq t_0$ (for some entry time t_0) *and* $dG/dt > 0$ (output is growing). In other words, the genius regime is the subset of the viability kernel in which the system is not merely surviving but actively expanding its intellectual output.

4.3 Existence of the genius regime

We now derive conditions under which the genius regime can be sustained.

Proposition 4.3 (Necessary viability conditions)**.** *If the genius regime (Definition 4.2) is sustained, then the following conditions must hold simultaneously at the boundary of the viability kernel:*

(V1) ***Plasticity–practice product exceeds decay:***

$$H \cdot \alpha(t) \cdot P_{\min} \cdot \sigma(E^*) \cdot S^* \cdot \phi(M_{\min}) > \delta_G \, G_{\min}. \tag{9}$$

That is, the combined strength of the hereditary endowment, current brain plasticity, minimal practice, minimal environmental support, minimal self-belief, and minimal motivation must be sufficient to overcome the natural decay of skills.

(V2) ***Self-belief sustains practice:***

$$\beta_P \, S^* \, M_{\min} + \eta_P \, E^* > \gamma_P \, P_{\min}. \tag{10}$$

That is, the combined effect of self-belief and motivation on practice, plus the facilitation provided by the environment, must be enough to counteract fatigue.

(V3) ***Output sustains environment (or environment is externally supported):***

$$\beta_E \, G_{\min} + \xi_E(t) > \gamma_E \, E^*. \tag{11}$$

That is, the Matthew effect (success attracting resources) plus any exogenous support must outpace environmental decay.

(V4) ***Progress sustains self-belief:***

$$\beta_S \cdot \frac{dG/dt}{G_{\min} + \epsilon} \cdot S^*(1 - S^*) > \gamma_S \, \psi\big(F(t)\big). \tag{12}$$

That is, the boost to self-belief from perceived progress must exceed the erosion caused by failure signals.

(V5) ***Novelty sustains motivation:***

$$\beta_M\,\nu(G_{\min}) > \gamma_M\,M_{\min}\,(1+\kappa\,R(t)). \tag{13}$$

That is, the motivational stimulation from encountering new ideas must exceed the combined effect of natural decay and external coercion.

Proof. The conditions are derived by evaluating the governing equations (1)–(5) at the corner of the viability kernel where each variable takes its minimum admissible value: $G = G_{\min}$, $P = P_{\min}$, $E = E^*$, $S = S^*$, $M = M_{\min}$. At this corner, maintaining viability requires that no variable's derivative is negative (otherwise the trajectory would exit $\mathcal{V}$). Substituting these boundary values into the right-hand sides of (1)–(5) and requiring $dG/dt > 0$, $dP/dt \geq 0$, $dE/dt \geq 0$, $dS/dt \geq 0$, $dM/dt \geq 0$ yields conditions (V1)–(V5). □

Remark 4.4 (Scope of the viability conditions). Conditions (V1)–(V5) are *necessary* for the genius regime to be sustained, but their sufficiency is not established here. In the full viability-theoretic framework of Aubin [63, 64], proving that a set is a viability kernel requires showing that for every boundary state, there exists an admissible trajectory remaining within the set. The corner-point analysis above addresses only the extremal configuration; the coupled, nonlinear dynamics may cause simultaneous approach of multiple variables to their boundaries, creating failure modes not captured by single-variable analysis. A full viability proof would require constructing a regulation map (in the sense of Aubin) or applying computational viability algorithms [65], which we leave to future work.

There is, however, a structural reason to expect that the gap between necessity and sufficiency is small. The multiplicative form of equation (1) implies that the product $P \cdot \sigma(E) \cdot S \cdot \phi(M)$ is *smallest* precisely when multiple variables approach their lower bounds simultaneously, since each factor is bounded below by its minimum and the product of several small quantities is smaller than any individual factor. The corner-point configuration $\mathbf{x}_{\min} = (G_{\min}, P_{\min}, E_{\min}, S_{\min}, M_{\min})$ therefore represents the hardest case for the viability condition to be satisfied. If all five inequalities hold at this extremal point, the system has strictly more "room" at any interior point of the viability kernel, suggesting that the necessary conditions are close to sufficient. A rigorous verification would require showing that the regulation map keeps all trajectories within the kernel, but the multiplicative structure provides an informal monotonicity argument that the boundary is the binding constraint.

Nevertheless, the necessary conditions are informative: if any one of (V1)–(V5) fails, the genius regime is *certainly* not sustainable, making them useful as diagnostic criteria.

Interpretation 4.5. Proposition 4.3 makes precise the widely held intuition that genius requires the *conjunction* of many conditions. It is not enough to have talent (high H); it is not enough to have a great environment (high E); it is not enough to believe in oneself (high S). All five conditions must be met simultaneously. The failure of any single condition causes the system to exit the viability kernel, potentially triggering a cascade of failures in other variables due to the coupling between equations.

5 Self-Belief as a Bifurcation Parameter

5.1 Two attractors: stagnation and growth

The most striking feature of the cognitive potential system is the existence of two qualitatively different long-term behaviours:

1. **The stagnation basin:** $G(t) \to G_{\text{low}}$, $P(t) \to 0$, $S(t) \to 0$, $M(t) \to M_{\text{low}}$. The individual ceases to practise, loses belief in their capacity, and output declines to a low level maintained only by residual environmental support.

2. **The growth trajectory:** $G(t) \to \infty$ (or, more realistically, toward a resource-limited plateau), $P(t) \to P_{\text{high}}$, $S(t) \to S_{\text{high}} < 1$, $M(t) \to M_{\text{high}}$. The individual engages in sustained practice, maintains strong self-belief, and produces growing intellectual output.

The variable that controls which attractor the system approaches is *self-belief* $S(t)$.

5.2 The bifurcation

A *bifurcation* is a qualitative change in the behaviour of a dynamical system that occurs when a parameter crosses a critical value. In our model, self-belief S acts as such a parameter because it appears multiplicatively in the equation for G and controls the equation for P.

Proposition 5.1 (Self-belief bifurcation)**.** *There exists a critical self-belief level $S_c \in (0, 1)$ such that:*

(a) *If $S(t) > S_c$ for a sustained period, the system converges to the growth trajectory.*

(b) *If $S(t) < S_c$ for a sustained period, the system converges to the stagnation basin.*

The critical value S_c is determined by the condition that the positive feedback loop (progress $\to$ self-belief $\to$ practice $\to$ progress) exactly balances the negative terms:

$$S_c = \frac{\gamma_P P_{\min} - \eta_P E}{\beta_P M}. \tag{14}$$

Conceptually, this defines a tipping point. Above this level, self-belief fuels practice, practice generates output, output reinforces self-belief, and the system spirals upward. Below this level, insufficient self-belief leads to reduced practice, reduced practice leads to declining output, declining output further erodes self-belief, and the system spirals downward.

Proof. From equation (2), at steady state we require $dP/dt = 0$, giving $P = (\beta_P SM + \eta_P E)/\gamma_P$. For $P \geq P_{\min}$, we need $S \geq (\gamma_P P_{\min} - \eta_P E)/(\beta_P M) = S_c$. When $S < S_c$, the steady-state practice falls below $P_{\min}$, which by Proposition 4.3 causes exit from the viability kernel. The multiplicative coupling between S and G in (1) then ensures that the decline in P propagates to G, which through (4) further reduces S, establishing the downward spiral. The symmetric argument establishes the upward spiral for $S > S_c$. $\square$

Remark 5.2 (Local approximation). The expression (14) for S_c is derived under the simplifying assumption that M and E are held constant while P equilibrates. In the full coupled system, M, E, and G are evolving simultaneously, so S_c is properly understood as a *local* bifurcation value—an instantaneous threshold that itself changes as the system evolves. The global dynamics may exhibit hysteresis: once the system has committed to the growth trajectory, returning to the stagnation basin may require S to fall below a value lower than S_c, because the accumulated G and E provide a buffer. This hysteresis is one of the model's more distinctive empirical predictions: it implies that individuals who have reached the growth regime should be *resilient* to temporary dips in self-belief, because accumulated output, environmental support, and motivational momentum act as a stabilizing reserve. The entry threshold into the growth regime would thus be higher than the exit threshold—a prediction testable through longitudinal studies tracking self-efficacy, practice behaviour, and performance across setbacks. A full global bifurcation analysis, accounting for the coupled dynamics on the slow manifold, would require numerical continuation methods and is left for future work.

Remark 5.3. Equation (14) reveals that S_c is *not fixed*: it depends on motivation M and environment E. A more supportive environment (higher E) lowers the critical threshold, making it easier to enter the growth regime. Higher intrinsic motivation (higher M) has the same effect. This formalizes the intuitive observation that a stimulating environment or a passionate interest can compensate for initially low self-confidence.

5.3 The autocatalytic interpretation

The positive feedback loop $S \to P \to G \to S$ is an example of an *autocatalytic cycle*—a process that accelerates itself. In chemistry, autocatalysis produces exponential growth until a limiting reactant is exhausted. In our model, the limiting "reactant" is the upper bound $S < 1$ imposed by the logistic term in (4), which prevents self-belief from growing without bound.

Interpretation 5.4 (Existential reading). The bifurcation at $S = S_c$ suggests—but does not formally entail—a parallel with the existentialist claim that genius is "not a gift but a choice." The system's long-term behaviour is determined not by the hereditary parameter H alone but by whether self-belief exceeds a threshold that is itself modifiable by environment and motivation. We emphasize that this is an *interpretive* observation, not a formal result: the model contains no mechanism for free will or agency. The initial condition $S(0)$ is simply given. What the model *does* show is that the relevant threshold is not fixed by biology but is a function of controllable environmental and motivational variables—a structural fact that is independent of one's philosophical stance on agency.

6 Neuroplasticity and the Learning Rule

6.1 Connecting the formalism to neural mechanisms

The abstract dynamical system of Section 3 acquires empirical content when its variables are connected to measurable neural processes. This section establishes those connections without claiming that the model is a detailed biophysical simulation—rather, it shows that the mathematical structure is *consistent with* known neuroscience and *motivated by* it.

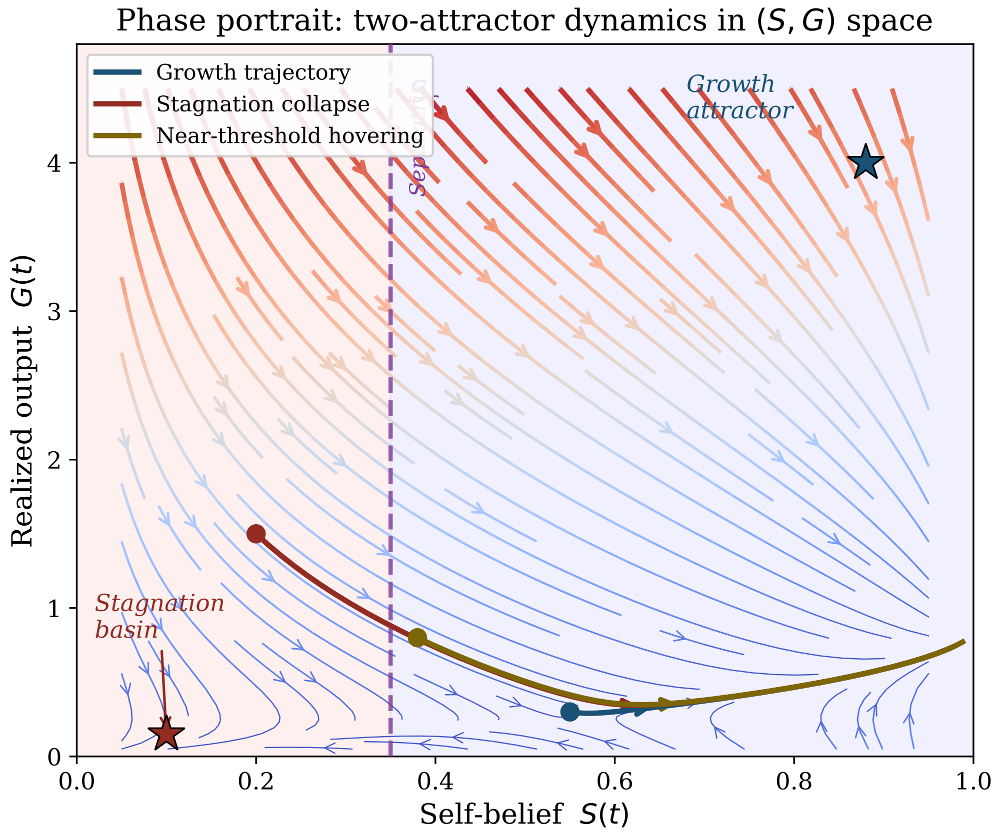

Figure 1: Phase portrait of the two-attractor system projected onto the (S, G) plane, with P, E, and M at representative values. The streamlines show the flow field; the dashed vertical line marks the separatrix at $S = S_c$. Three sample trajectories are shown: one entering the growth attractor (blue), one collapsing into the stagnation basin (red), and one hovering near the threshold before committing (gold). Stars mark the approximate locations of the two attractors. The red-shaded region ($S < S_c$) corresponds to inevitable stagnation; the blue-shaded region ($S > S_c$) permits entry into the growth regime.

6.2 The adaptive network model

At the neural level, learning corresponds to changes in the strength of connections (synapses) between neurons. We denote the synaptic weight between neuron i and neuron j as w_{ij}. The change in this weight can be modelled by a learning rule that depends on *prediction error*—the discrepancy between what the brain expected and what actually occurred.

Definition 6.1 (Prediction-error learning rule)**.** Let $\hat{y}$ denote the brain's prediction and y denote the actual outcome. The **prediction error** is $e = y - \hat{y}$. The synaptic weight w_{ij} is updated according to:

$$\Delta w_{ij} = \alpha(t) \cdot e \cdot x_i \cdot x_j, \tag{15}$$

where x_i and x_j are the activation levels of neurons i and j, and $\alpha(t)$ is the neuroplasticity coefficient from equation (6).

The intuition is straightforward: when the brain's prediction is wrong, the connections between the neurons that were active at the time of the error are strengthened or weakened in proportion to how wrong the prediction was and how active the neurons were. The plasticity coefficient $\alpha(t)$ controls how rapidly this adjustment occurs.

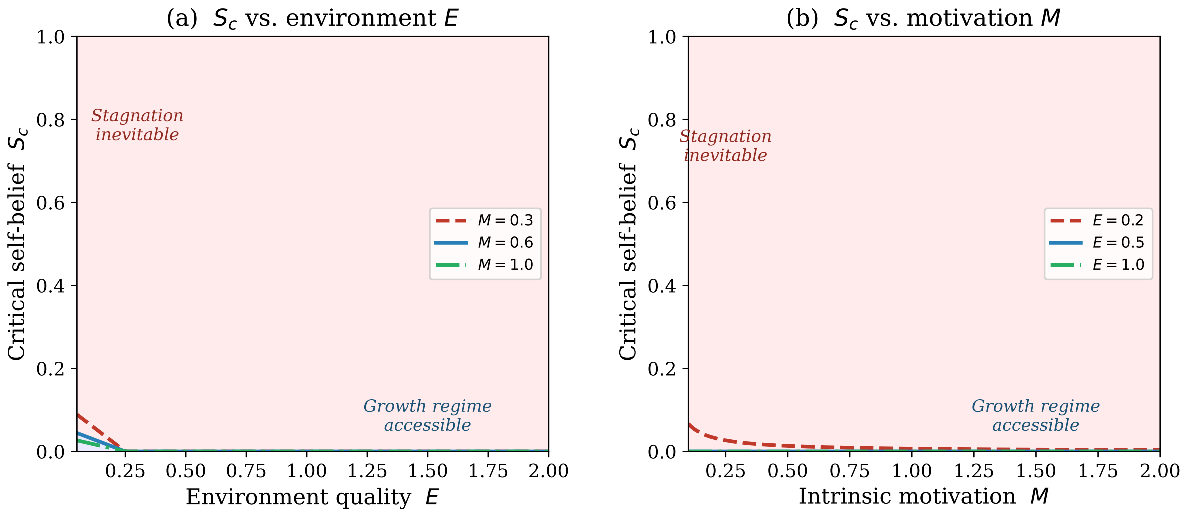

Figure 2: Bifurcation diagram showing how the critical self-belief threshold S_c depends on (a) environmental quality E and (b) intrinsic motivation M. Higher values of either variable lower the threshold, expanding the region in which the growth regime is accessible. The red-shaded region above the curve represents parameter combinations for which stagnation is inevitable regardless of initial self-belief; the blue-shaded region below represents conditions under which the growth trajectory can be reached.

This rule is a generalization of Hebbian learning [13] ("neurons that fire together wire together") with the crucial addition of an error signal. It corresponds closely to the dopaminergic prediction-error signal discovered by Schultz, Dayan, and Montague [18], where dopamine neurons fire when outcomes are better than expected and suppress firing when outcomes are worse than expected, providing a biological implementation of temporal-difference learning [19]. Long-term potentiation (LTP), the cellular mechanism underlying synaptic strengthening, was first demonstrated by Bliss and Lømo [14] and provides the neurophysiological substrate for this weight-update process.

6.3 Errors as drivers of growth

Proposition 6.2 (Error-driven improvement)**.** *In the prediction-error learning framework, the rate of improvement of a cognitive skill is proportional to the frequency and magnitude of* corrected *errors. Specifically, let $L(t)$ denote a loss function measuring the discrepancy between desired and actual performance. Then:*

$$\frac{dL}{dt} = -\alpha(t) \cdot f_{err}(t) \cdot |\bar{e}(t)|, \tag{16}$$

where $f_{err}(t)$ is the frequency of error-correction events per unit time and $|\bar{e}(t)|$ is the average magnitude of prediction errors.

Intuitively, skills improve faster when the individual makes mistakes *and corrects them*, and when the brain is plastic enough to adjust its connections in response. This provides the neural basis for the concept of "deliberate practice"—practice that is specifically designed to push the individual into the zone where errors are frequent and informative.

Remark 6.3. The plasticity parameter $\alpha(t)$ plays a dual role. If α is too low, the brain cannot adjust quickly enough and the individual becomes rigid, unable to learn new skills or abandon incorrect beliefs. If α is too high, the brain over-adjusts to every new piece of information, leading to instability—the individual cannot retain learned skills because each new experience overwrites previous learning. The optimal value of α represents a balance between stability and adaptability, and the fact that α declines with age (equation 6) explains both the learning advantage of youth and the accumulated wisdom (stability of learned patterns) of maturity.

7 Historical Convergence of Discoveries

7.1 The phenomenon

One of the most striking patterns in the history of ideas is *simultaneous independent discovery*: Newton and Leibniz both developing calculus, Darwin and Wallace both arriving at evolution by natural selection, multiple physicists converging on special relativity in the early 1900s. These are not coincidences; they are manifestations of a deeper structural phenomenon. Ogburn and Thomas [51] documented 148 such instances, and Merton [49] argued that "multiples" are the rule rather than the exception in science, reflecting the structural maturity of the knowledge base rather than individual genius alone.

7.2 The knowledge-graph percolation model

We model the total body of human knowledge at time t as a graph $\mathcal{K}(t)$, where nodes represent concepts, facts, and methods, and edges represent known connections between them. A *discovery* corresponds to the identification of a new edge—a connection between two previously unlinked concepts. This model is presented as a *complementary* framework to the individual-level ODE system of Sections 3–5; the two are coupled through the historical context parameter $C(t)$ (Definition 3.4), which we now identify with the knowledge-graph density: $C(t) \propto \rho(t)$. An individual's realized output $G(t)$ contributes to the graph by adding edges (publications, inventions, artistic works), while the graph's density determines the accessibility of concepts that feed into the individual's environmental variable $E(t)$.

Definition 7.1 (Knowledge graph)**.** The **knowledge graph** $\mathcal{K}(t) = (N(t), L(t))$ consists of a set of nodes $N(t)$ (concepts) and a set of links $L(t)$ (known connections). The **density** of the graph is $\rho(t) = |L(t)|/\binom{|N(t)|}{2}$, the ratio of actual connections to possible connections. We model $\mathcal{K}(t)$ as an evolving random graph in the Erdős–Rényi family $G(n, p)$, where $n = |N(t)|$ and $p = \rho(t)$, with edges added according to the discovery process defined below.

Definition 7.2 (Discovery probability)**.** Let $d = (n_1, n_2)$ be a potential discovery (an edge not yet in $L(t)$) connecting concepts n_1 and n_2. The **probability that an agent with realized output $G_i(t)$ discovers** d in a time interval $[t, t + \Delta t]$ is:

$$\Pr(d \text{ discovered by agent } i \text{ in } [t, t+\Delta t]) = 1 - \exp\big(-\mu \cdot G_i(t) \cdot A(n_1, t) \cdot A(n_2, t) \cdot \Delta t\big), \quad (17)$$

where $A(n, t)$ is the **accessibility** of concept n at time t—a measure of how widely known and well-understood that concept is—$\mu > 0$ is a coupling constant, and $G_i(t)$ is the agent's realized output from the individual-level model, providing the formal coupling between the two frameworks. Note that the subscript i implicitly extends the model to a multi-agent setting: the knowledge graph $\mathcal{K}(t)$ is a shared, population-level object, while each agent i possesses an individual state vector $(G_i, P_i, E_i, S_i, M_i)$ governed by the equations of Section 3. This paper does not develop the multi-agent dynamics (peer effects, competition, mentoring); the subscript i should be read as indexing independent agents who interact only through their contributions to the shared knowledge graph.

Intuitively, a discovery becomes increasingly likely as both of the concepts it connects become widely known *and* as the individual agent's cognitive capacity grows. When the knowledge base reaches a critical density, the accessibility of many concepts exceeds a threshold, and the probability of the connecting discovery jumps sharply for multiple agents simultaneously.

[Convergence threshold] Under the Erdős–Rényi graph model for $\mathcal{K}(t)$, there exists a critical knowledge density ρ^* such that for $\rho(t) < \rho^*$, the expected number of independent co-discoveries per century is near zero, while for $\rho(t) > \rho^*$, it increases rapidly. Specifically, we conjecture:

$$\mathbb{E}[\text{co-discoveries per century}] \sim \begin{cases} O(1) & \text{if } \rho(t) \ll \rho^*, \\ \Theta\big((\rho(t) - \rho^*)^\gamma\big) & \text{if } \rho(t) > \rho^*, \end{cases} \tag{18}$$

where $\gamma > 0$ is a critical exponent. In standard Erdős–Rényi percolation, the critical threshold is $\rho^* = 1/n$ and $\gamma = 1$; the extent to which these values carry over to the accessibility-weighted model of equation (17) remains an open question requiring separate analysis.

Remark 7.3. We label this a conjecture rather than a proposition because the accessibility-weighted discovery probability (17) introduces correlations between edge-formation events that break the independence assumptions of classical Erdős–Rényi theory. Establishing the precise critical exponent would require either a mean-field analysis of the correlated percolation process or numerical simulation of the evolving graph. The qualitative prediction—that a threshold exists above which co-discoveries become frequent—is robust across graph ensembles and is supported by the historical data compiled by Ogburn and Thomas [51] and Merton [49].

Interpretation 7.4. This model suggests that simultaneous independent discovery reflects a structural property of the knowledge base rather than coincidence. When the knowledge graph is sparse, only individuals with exceptional access to diverse areas of knowledge (high personal G and E) can bridge the gaps. When the graph becomes dense, the bridges are so short that multiple investigators are likely to find them independently within a short time. The individual genius matters not for *whether* the discovery is made, but for *how* it is articulated, developed, and communicated.

8 Artificial Intelligence as Cognitive Extension

8.1 The extended cognitive system

We now incorporate artificial intelligence into the dynamical framework. The key idea is that AI does not replace the human cognitive system but *extends* it, adding computational capacity that is limited by the bandwidth of the interface between the two systems. This view is grounded in Licklider's [54] concept of human–computer symbiosis, Engelbart's [55] framework for augmenting human intellect, and the extended mind thesis of Clark and Chalmers [59], which argues that external devices participating in cognitive processes should be considered part of the cognitive system itself.

Definition 8.1 (Extended cognitive capacity). The **total cognitive capacity** of the human–AI system is:

$$K_{\text{total}}(t) = K_{\text{brain}}(t) + \beta(t) \cdot K_{\text{AI}}(t), \tag{19}$$

where:

- $K_{\text{brain}}(t) = G(t)$ is the human's realized cognitive output.
- $K_{\text{AI}}(t)$ is the computational capacity of the AI system available to the individual.
- $\beta(t) \in [0, 1]$ is the **integration coefficient**, representing the fraction of the AI's capacity that the individual can effectively access and utilize.

The integration coefficient β depends on the **bandwidth** $B(t)$ of the human–AI interface:

Definition 8.2 (Interface bandwidth). The **bandwidth** $B(t)$ is the rate at which information can flow between the human brain and the AI system, measured in bits per second. The integration coefficient is:

$$\beta(t) = 1 - e^{-B(t)/B^*}, \tag{20}$$

where B^* is the characteristic bandwidth at which half of the AI's capacity becomes accessible.

The interpretation is direct: when the interface is very slow (typing text queries, reading answers on a screen), B is small and β is close to zero—the AI's vast computational power is largely inaccessible because the channel between it and the brain is too narrow. Card, Moran, and Newell [56] estimated the human deliberate information-processing rate at approximately 7–10 bits per second; with current keyboard-and-screen interfaces, B is at most 50 bits per second. As the interface improves (voice interaction, and eventually neural interfaces), B increases and β approaches 1, meaning nearly all of the AI's capacity becomes available.

8.2 The bottleneck equation

Proposition 8.3 (Cognitive bottleneck). *The rate of growth of realized output in the extended system is:*

$$\frac{dG_{ext}}{dt} = \frac{dG}{dt} + \beta(t) \cdot \min\big(K_{AI}(t),\ B(t)\big) \cdot \sigma(E) \cdot S, \tag{21}$$

where $\min(K_{AI}, B)$ *reflects the fact that the effective augmentation is limited by the lesser of the AI's capacity and the bandwidth.*

The core insight is that no matter how powerful the AI is, the benefit to the individual is capped by the bandwidth of the interface. This is the *bottleneck*: the narrow channel through which the AI's vast capabilities must pass before they can augment human thought. With current technology (keyboard and screen), B is approximately 50 bits per second. Theoretical neural interfaces could increase this by orders of magnitude, fundamentally changing the dynamics of the system.

Remark 8.4. Proposition 8.3 also makes clear that the AI's contribution is modulated by $\sigma(E) \cdot S$: even with a perfect interface, the individual needs an adequate environment and sufficient self-belief to benefit from the AI. An individual with $S = 0$ (complete loss of self-belief) gains nothing from AI augmentation, because they do not engage with the system. This is a formal expression of the book's insight that "technology does not replace the activity of the mind."

9 Creativity as Divergent–Convergent Balance

9.1 Two modes of thought

Creativity involves two complementary cognitive processes. *Divergent thinking*—introduced by Guilford [37, 38] and operationalized in the Torrance Tests [39]—generates multiple candidate ideas, while *convergent thinking* evaluates, filters, and selects the most promising ones. Neither process alone produces creative output: divergence without convergence yields chaos, and convergence without divergence yields sterility. This two-stage model—generate, then select—was formalized as "blind variation and selective retention" (BVSR) by Campbell [42] and Simonton [10], and is supported by the associative theory of Mednick [40] and the network models of Kenett et al. [41].

Definition 9.1 (Creativity as balance)**.** Let $D(t)$ denote the **divergent coefficient**—the rate at which the cognitive system generates alternative hypotheses, ideas, or solutions per unit time. Let $F_c(t)$ denote the **convergent filter**—the stringency with which generated ideas are evaluated and rejected. The **effective creativity** is:

$$\mathcal{C}(t) = D(t) \cdot \big(1 - F_c(t)\big) \cdot Q\big(D(t), F_c(t)\big), \tag{22}$$

where Q is a **quality function** that captures the average quality of ideas that pass the filter.

That is, creativity depends on generating many ideas (D high) while filtering out the weak ones (F_c not too low), but with the filter not so strict that it rejects everything (F_c not too high).

Proposition 9.2 (Optimal creativity)**.** *Under reasonable assumptions on Q (specifically, that Q increases with F_c and decreases with D beyond a saturation point), the effective creativity $\mathcal{C}(t)$ is maximized at an interior point (D^*, F_c^*) where:*

$$\left.\frac{\partial \mathcal{C}}{\partial D}\right|_{D^*, F_c^*} = 0, \qquad \left.\frac{\partial \mathcal{C}}{\partial F_c}\right|_{D^*, F_c^*} = 0. \tag{23}$$

In words: the genius regime corresponds to an *optimal balance* between generating many ideas and rigorously filtering them. Too much divergence (high D, low F_c) produces a flood of weak ideas—noise rather than insight. Too much convergence (low D, high F_c) produces a trickle of safe, conventional ideas—competence without originality. The genius operates at the point where the two processes are calibrated to produce the maximum flow of high-quality, original ideas.

Remark 9.3 (Integration with the governing equations). The effective creativity $\mathcal{C}(t)$ enters the main dynamical system as a multiplier on the growth equation (1): when the individual operates near the optimal creativity point (D^*, F_c^*), the rate of output growth is enhanced. Specifically, we can extend equation (1) to

$$\frac{dG}{dt} = H \cdot \alpha(t) \cdot P(t) \cdot \sigma(E) \cdot S \cdot \phi(M) \cdot \big(1 + \lambda_{\mathcal{C}}\, \mathcal{C}(t)\big) - \delta_G\, G, \tag{24}$$

where $\lambda_{\mathcal{C}} \geq 0$ is a creativity amplification parameter. The additive $(1 + \lambda_{\mathcal{C}}\, \mathcal{C})$ form—rather than a purely multiplicative factor—is a deliberate modelling choice reflecting the distinction between *output* and *creative output*. The multiplicative factors in the base equation (H, α, P, $\sigma(E)$, S, $\phi(M)$) are all *necessary*: any one at zero suppresses growth entirely. Creativity, by contrast, is an *optional amplifier*: an individual can produce high-quality non-creative output (skilled performance, thorough analysis, expert synthesis) with $\mathcal{C} = 0$, but creative breakthroughs require $\mathcal{C} > 0$ to augment the baseline growth rate. If one wished to model a regime in which creativity is itself necessary—where only novel contributions count as "genius output"—the appropriate extension would be to redefine G as creative output and let $\mathcal{C}$ enter multiplicatively. The additive form thus represents the more general case; the multiplicative case is recovered when $\lambda_{\mathcal{C}}\, \mathcal{C}$ dominates the baseline. At the baseline $\mathcal{C} = 0$ (no creative output beyond routine work), the equation reduces to (1). At high creativity, the growth rate is amplified. The divergent coefficient D is itself a function of intrinsic motivation M and the novelty of the environment [36], while the convergent filter F_c depends on practice-acquired expertise P and analytical skill, providing the bidirectional coupling between the creativity module and the core state variables.

Remark 9.4 (Neurochemical interpretation). The D/F_c balance maps onto the neurochemical balance described in the neuroscience of creativity. Divergent thinking is associated with the default mode network [20, 21] and facilitated by dopaminergic tone, while convergent filtering is associated with executive control networks and mediated by prefrontal cortical activity. Beaty et al. [22] showed that highly creative individuals exhibit stronger functional connectivity between these two networks, suggesting that creativity depends on their coordinated interaction. Martindale [44] proposed a cortical-arousal model in which low arousal facilitates divergent generation while high arousal facilitates convergent evaluation—a dynamic that maps directly onto the (D, F_c) pair.

10 Curiosity as Optimal Information Search

Curiosity—the drive to seek new information—can be formalized as an optimal search strategy over the individual's model of the world.

Definition 10.1 (Curiosity function)**.** Let $\mathcal{W}(t)$ denote the individual's internal model of the world (we use $\mathcal{W}$ rather than $\mathcal{M}$ to avoid confusion with the intrinsic motivation variable $M(t)$). Let $\mathcal{H}(\mathcal{W})$ denote the entropy (uncertainty) of this model. The **curiosity function** is:

$$\mathrm{Cur}(a, t) = \mathbb{E}[\Delta\mathcal{H}(\mathcal{W}) \,|\, \text{action } a]\,, \tag{25}$$

where the expectation is over possible outcomes of action a.

Intuitively, the curiosity value of an action is the expected amount by which that action will *change* (typically reduce) the uncertainty in the individual's understanding of the world. A highly curious agent selects actions that maximize this quantity.

Proposition 10.2 (Optimal curiosity)**.** *An agent that maximizes $Cur(a,t)$ subject to the constraint that total entropy does not exceed a cognitive overload threshold $\mathcal{H}_{\max}$ implements the following strategy:*

$$a^*(t) = \arg\max_a \; Cur(a,t) \quad \textit{subject to} \quad \mathcal{H}(\mathcal{W}(t)) \leq \mathcal{H}_{\max}. \tag{26}$$

In words: the optimally curious agent seeks the most informative actions but avoids overwhelming itself with more uncertainty than it can process. This formalization captures two observations from the book: (1) that curiosity is the "natural fuel" of intellectual development, and (2) that raw information without understanding is a "useless burden"—the overload constraint $\mathcal{H}_{\max}$ prevents the agent from accumulating information faster than it can be integrated into knowledge.

Remark 10.3 (Integration with the governing equations)*.* The curiosity function connects to the core dynamical system through the novelty function $\nu(G)$ in the motivation equation (5). In the governing equations (Section 3), $\nu(G)$ was introduced as an assumed concave, increasing function without specifying its microfoundation. The curiosity formalization provides such a microfoundation: we *interpret* the previously assumed $\nu(G)$ as $\nu(G) = \max_a \mathrm{Cur}(a,t)$ evaluated at the current knowledge level. This is not a redefinition but a derivation of functional form from first principles: as G grows, the individual's world-model $\mathcal{W}$ expands, opening new frontiers of uncertainty that sustain curiosity and thereby sustain motivation M. The concavity of ν follows from the overload constraint—at high G, the marginal curiosity value of further exploration diminishes because the agent approaches $\mathcal{H}_{\max}$. The overload constraint thus also provides a formal mechanism for the motivational collapse that can occur when the individual is overwhelmed by complexity—corresponding to the transition from the growth trajectory to the stagnation basin when M drops below $M_{\min}$.

11 The Integrated Model

Figure 6 summarizes the complete dynamical system, showing all state variables, parameters, feedback loops, and the AI extension module.

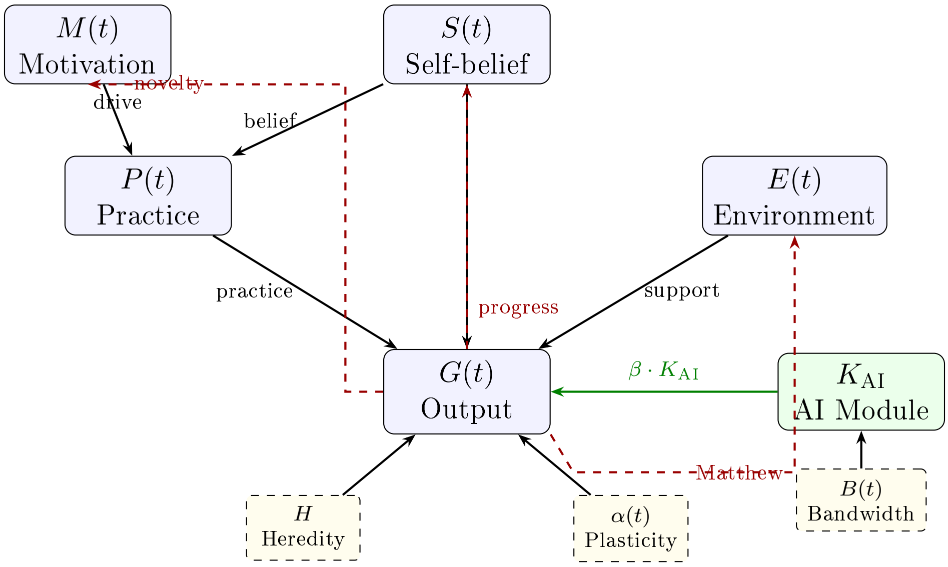

Figure 6: Architecture of the cognitive potential dynamical system. Solid arrows represent direct causal influences; dashed red arrows represent feedback loops. The green arrow represents the AI extension module, whose contribution is limited by the bandwidth $B(t)$.

12 Robustness, Sensitivity, and Limitations

A critical question for any mathematical model is the extent to which its qualitative predictions depend on the specific functional forms chosen. This section addresses the reviewer's natural concern: would the two-attractor structure, the self-belief bifurcation, and the multiplicative coupling survive under alternative modelling choices?

12.1 Structural robustness of the two-attractor result

The existence of two qualitatively different long-term behaviours—the stagnation basin and the growth trajectory—is the paper's most important qualitative prediction. We argue that this result is *structurally robust*: it survives under a broad class of alternative functional forms, because it depends on two features that are generic rather than specific to our particular equations:

(i) **Positive feedback through self-belief.** Any model in which self-belief promotes practice, practice promotes output, and output promotes self-belief will exhibit positive feedback. The specific forms of equations (2) and (4) can be altered—for instance, S could enter P through a power law rather than linearly, or the self-belief update could use a different bounded growth function instead of the logistic—without eliminating the feedback loop. What matters is the *sign structure* of the Jacobian matrix: $\partial(dP/dt)/\partial S > 0$, $\partial(dG/dt)/\partial P > 0$, $\partial(dS/dt)/\partial(dG/dt) > 0$.

(ii) **Decay in the absence of input.** Any model in which skills erode without practice ($\delta_G > 0$) and self-belief erodes under failure ($\gamma_S > 0$) will have a low-output attractor. The stagnation basin is structurally guaranteed by the presence of these decay terms, regardless of their precise functional form.

The bifurcation between the two attractors is also generic. In any system with a positive feedback loop competing against decay, there exists a threshold below which decay dominates and above which growth dominates. The specific expression (14) for S_c would change under alternative functional forms, but the *existence* of a threshold would not.

12.2 Sensitivity to functional form choices

Certain features of the model *are* sensitive to the specific choices made:

- **Motivation activation function.** We chose a Michaelis–Menten form $\phi(M) = M/(M + K_M)$ for mathematical convenience and because it captures saturating returns. A sigmoidal function $\phi(M) = 1/(1+e^{-k_M(M-M^*)})$ would introduce a threshold effect in motivation, potentially creating a third attractor (a "plateau" regime where output is positive but creativity is suppressed). The qualitative two-attractor result is preserved under either choice, but the sigmoidal form admits richer dynamics.

- **Self-belief dynamics.** The logistic term $S(1 - S)$ was chosen to bound $S \in [0, 1]$, which is natural for a probability-like variable. Alternative bounded growth models (e.g., Gompertz growth) would yield similar qualitative results but different quantitative predictions for the speed of self-belief change.

- **Environmental equation.** We acknowledged (and the reviewer rightly noted) that the environmental equation (3) is linear, while the others are nonlinear. A more realistic formulation would include both a saturation effect (environments cannot improve indefinitely) and a threshold effect (below a minimum, the environment actively suppresses development):

$$\frac{dE}{dt} = \beta_E \frac{G(t)}{G(t) + K_E} + \xi_E(t) - \gamma_E E(t) - \delta_E \max(0, E_{\text{crit}} - E(t)), \qquad (27)$$

  where K_E is a half-saturation constant for the Matthew effect and δ_E represents the suppressive effect of environments below a critical quality E_{crit}. This extension introduces a possible third equilibrium (an "environmental poverty trap") that merits clarification of its relationship to the paper's central two-attractor structure. The poverty trap is best understood as a *pre-condition* rather than a modification of the stagnation/growth dynamics: it describes an absorbing state for individuals whose environmental quality E is so low that it cannot reach the viability threshold $E_{\min}$ regardless of the individual's effort, self-belief, or talent. For individuals above E_{crit}, the two-attractor dynamics described in Sections 4–5 operate as before. The poverty trap thus partitions the population into two regimes: those for whom the genius question is meaningful (above E_{crit}) and those for whom structural environmental barriers preclude entry into the viability kernel entirely. A full analysis of the three-equilibrium dynamics—including the possibility that policy interventions can shift E_{crit} and thereby enlarge the accessible population—is left for future work.

12.3 What this model cannot do

We conclude this section with an explicit statement of the model's limitations, in the spirit of intellectual honesty that the framework itself demands:

(i) **No quantitative predictions.** The model is not calibrated to empirical data, and its parameters are unspecified. It cannot predict, for instance, the number of hours of practice required to enter the genius regime, or the effect size of a particular environmental intervention. Empirical calibration would require longitudinal data on all five state variables simultaneously, which currently does not exist in any single dataset.

(ii) **No mechanism for individual differences in functional form.** The model assumes all individuals are governed by the same system of equations with different parameter values. In reality, the *functional form* of the equations may differ across individuals (e.g., some people may have sigmoidal rather than saturating motivation dynamics). Structural variation across individuals is a major unsolved problem in mathematical psychology.

(iii) **No stochasticity.** The model is deterministic. Real cognitive trajectories are subject to random perturbations—chance encounters, illness, serendipitous discoveries, economic shocks. A stochastic extension (replacing the ODEs with stochastic differential equations) would be needed to model the role of luck in cognitive development, and would likely show that the basins of attraction for the two regimes have fuzzy boundaries.

(iv) **No social interaction dynamics.** The model treats the individual in isolation, with social effects captured only through the aggregate environment variable $E(t)$. A multi-agent extension, in which individuals influence each other's self-belief and motivation, would be needed to capture phenomena such as peer effects, mentoring, and the social construction of genius.

(v) **Epistemological status.** The model is a *conceptual framework*—a mathematical language for organizing and connecting empirical findings—not a validated theory. Its value lies in generating structural hypotheses (e.g., that self-belief controls a bifurcation, that the genius regime requires conjunction of conditions, that AI augmentation is bandwidth-limited) that can guide empirical research and intervention design. We do not claim that the model has been "proven" or that its predictions have been empirically verified.

13 Location Within the Unified Structural Theory of Complex Systems

13.1 A theory that emerged from its parts

This paper is part of a broader research programme—the Unified Structural Theory of Complex Systems [69]—that seeks to identify formal laws governing the behaviour of complex systems across physical, cognitive, and social domains. The development of this programme has followed an unusual trajectory that deserves brief explanation.

The programme did not begin with a grand plan or a predetermined theoretical architecture. Rather, it began with individual investigations into specific problems: the structural basis of self-organization [70], the dynamics of persistence in changing environments [71], the predictive processing architecture of cognition [72], the formal structure of extractive oscillators in social systems [73], and applications ranging from binary star formation [74] to clinical decision-making [75]. Each paper was written as a self-contained contribution to its respective field.

It was only after a substantial number of these papers had been completed that a remarkable pattern became apparent: the same small set of structural principles—persistence, viability, closure, boundary maintenance, and resilience—kept reappearing across domains that had no obvious connection to one another. The laws governing the stability of a biological cell turned out to be structurally isomorphic to the laws governing the stability of a social institution; the predictive architecture that brains use to process sensory information was derivable from the same constraint equations that describe self-sufficient dynamical systems; the conditions under which a binary star system remains bound were analogous to the conditions under which a cooperative relationship remains viable.

This convergence was not imposed by the author but *discovered* in the mathematical structure of the individual results. The Unified Structural Theory emerged as the recognition that these recurring patterns may represent instances of the same formal constraints operating at different scales and on different substrates. Whether the correspondences are genuine structural isomorphisms or productive analogies that happen to share mathematical form is a question that each individual paper addresses for its own domain; the cross-domain claim is offered as a research programme, not as a settled conclusion.

13.2 The six-layer architecture

The theory, as it has crystallized, comprises six layers:

Part I. **Meta-Theory.** Foundational results about the limits and structure of formalization itself: the law of scale-specific principles (no single theory can encompass all scales), the formalization laws (no single formalization captures all operationally relevant properties), definition-dependent provability, imperative uncertainty, and the limit to negation [76–80].

Part II. **Dynamical Core.** The structural laws that any persistent complex system must satisfy: the persistence triad (closure, boundary maintenance, resilience), the four laws of self-organization (cooperation, viability, interference, observability), the constraint–autonomy compatibility law, and the viability mismatch law [70, 71, 81, 82].

Part III. **Cognition.** The architecture of cognitive systems derived (not hypothesized) from the dynamical core: predictive processing as the only viable architecture under energy and latency constraints, the structural distortion principle (bounded systems must distort representations), representational isolation, atemporal memory, and models of mental disintegration [72, 83–86].

Part IV. **Social Systems.** Formal models of social phenomena derived from the same structural laws: extractive oscillators (a substrate-independent model capturing narcissistic dynamics, addiction, and algorithmic dependence), conflict as phase transition, the asymmetry of totalizing ideals, and frameworks for deception and autonomy suppression [73, 87–90].

Part V. **Cross-Domain Applications.** The structural laws applied to specific domains: astrophysics (binary star formation, dormant neutron stars), cosmology (Bayesian model comparison), information theory (biospheric complexity, local entropy inversion), AI and civilization (the inward turn hypothesis for the Fermi paradox, the stimulus problem in post-scarcity environments), and healthcare (clinical discontinuity) [74, 75, 91–96].

Part VI. Synthesis. The deepest result: differentiation as the ontological condition of actualization, the principle of optimal coherence, and the informational preconditions of meaning [97–99].

The central thesis of the unified theory is: *Persistence, not truth, is the primary organizing principle of complex systems. Truth-seeking, consciousness, social organization, and scientific inquiry are derivative phenomena—strategies that systems have evolved to maintain structural viability.*

13.3 Where this paper fits

The present paper fills a specific gap in this architecture. The dynamical core (Part II) provides the abstract laws of viability, persistence, and self-organization. The cognition layer (Part III) derives the architecture of cognitive systems from these laws. But neither layer addresses the question of how a cognitive system can *optimize its own trajectory*—how it can move from ordinary functioning to the regime of maximal intellectual output that we call genius.

This paper bridges Parts II, III, and IV by showing that the realization of cognitive potential is itself a viability problem: the genius regime is a viability kernel in a dynamical system whose variables include neural plasticity, deliberate practice, environmental quality, self-belief, and intrinsic motivation. The self-belief bifurcation (Section 5) connects to the extractive oscillator framework of Part IV (the stagnation basin is a degenerate case of an extractive cycle, while the growth trajectory is its constructive inverse). The AI augmentation model (Section 8) connects to the applications in Part V (AI-extended communication, the stimulus problem). And the convergence theorem (Section 7) is an application of the structural non-neutrality principle from Part II.

Table 2 summarizes the correspondences between this paper's constructs and the broader theory.

Table 2: Correspondences between constructs in this paper and the Unified Structural Theory.

This paper	**Unified Theory construct**	**Layer**
Viability kernel $\mathcal{V}$	Viability set	Part II
Self-belief bifurcation	Attractor structure	Part II
Stagnation basin	Extractive oscillator (degenerate)	Part IV
Growth trajectory	Generative oscillator	Part III
Neuroplasticity $\alpha(t)$	Predictive processing update rule	Part III
Error-driven learning	Structural distortion correction	Part III
Knowledge-graph convergence	Structural non-neutrality	Part II
AI bandwidth $B(t)$	Communication channel capacity	Part V
Creativity balance D/F_c	Divergent–convergent processing	Part III
Curiosity function	Optimal information search	Part III
Existential interpretation	Differentiation $\to$ actualization	Part VI

14 From Theory to Practice: Application Scenarios

The mathematical framework developed in this paper is not merely an abstract exercise. Its variables, thresholds, and feedback loops map onto concrete intervention points that can, in principle, be operationalized as practical tools. This section sketches five application scenarios that illustrate how the model's structure could inform technology design. We stress that these are *design concepts* motivated by the formalism, not validated products; each would require substantial engineering and empirical evaluation.

14.1 Personal potential navigator

The governing equations (Section 3) identify five state variables that must simultaneously remain above threshold for the genius regime to be sustained. An AI-assisted coaching application could monitor proxies for these variables—self-reported self-belief S, logged practice hours P, environmental engagement E, and motivational indicators M—and identify the current *bottleneck*: the variable closest to its viability boundary. Rather than generic advice, the system would provide structurally informed recommendations: "Your practice intensity is high, but self-belief is declining. The model predicts that sustained practice under falling S will lead to exit from the growth regime (Proposition 5.1). Focus on consolidating small, verifiable achievements to restore S above S_c." The adaptive complexity of learning materials could be calibrated to keep the user in the zone where the error-driven learning rule (Section 6) is maximally effective—neither so easy that $\Delta w_{ij} \approx 0$ nor so difficult that motivational overload (the $\mathcal{H}_{\max}$ constraint of Section 10) triggers collapse.

14.2 Environmental ecosystem analyzer

The environmental variable $E(t)$ is governed by equation (3), in which the Matthew effect ($\beta_E G$) and exogenous support (ξ_E) compete against decay ($\gamma_E E$). An application could audit an individual's contact network, institutional affiliations, and available cognitive tools to estimate E and its trajectory. Recommendations would target the specific environmental deficits the model identifies: "Your E score is below the viability threshold $E_{\min}$. The model suggests increasing ξ_E (exogenous support) through professional community membership or AI tool adoption, since the Matthew-effect term is currently too weak to sustain E autonomously." The nonlinear environmental extension (equation 27) further identifies the poverty-trap threshold E_{crit}, below which no amount of individual effort suffices—alerting the user (or a policy designer) to structural barriers that require systemic intervention.

14.3 Cognitive bandwidth trainer

The bottleneck equation (Proposition 8.3) formalizes the constraint that AI augmentation is limited by interface bandwidth $B(t)$. An application could measure the speed and quality of a user's interaction with AI systems—query formulation, result interpretation, iterative refinement—and provide targeted training to increase effective B. As Figure 5 illustrates, even modest improvements in bandwidth (from ∼50 bps to ∼200 bps via voice interfaces) produce disproportionate gains in the effective coupling β. The application would also help the user strategically offload cognitive tasks to AI, freeing biological resources for the creative synthesis that the model locates in the divergent–convergent balance (Section 9).

14.4 Viability dashboard with biometric integration

Integration with wearable sensors (sleep quality, heart-rate variability, activity patterns) could provide real-time proxies for the neuroplasticity coefficient $\alpha(t)$ and practice capacity $P(t)$. The dashboard would track the system's proximity to the viability boundary, issuing early warnings when physiological indicators suggest that the trajectory is approaching the stagnation basin: "Your recovery metrics indicate declining $\alpha(t)$. The viability condition (V1) is at risk. Preventive rest is recommended to avoid crossing the bifurcation threshold." The hysteresis prediction of Remark 5.2 is particularly relevant here: the model predicts that once the system has entered the stagnation basin, recovery requires S to rise *above* the original entry threshold, making prevention substantially more efficient than remediation.

14.5 Structurally informed mentoring platform

The multiplicative structure of equation (1) implies that the most effective intervention targets whichever factor is currently smallest, since the marginal return of improving the binding constraint exceeds that of improving any other variable. A mentoring platform could match mentees with mentors whose expertise addresses the specific bottleneck: a motivational mentor for low M, a technical expert for inadequate P, or an environmental connector for deficient E. The system-level matching algorithm would optimize not for generic "mentoring quality" but for the expected increase in the product $P \cdot \sigma(E) \cdot S \cdot \phi(M)$, targeting the factor with the highest marginal return given the mentee's current state vector.

The common thread across these scenarios is a shift from intuition-driven self-development to *structurally informed* intervention, where each action is grounded in the individual's current position within the dynamical system. We emphasize the limitations noted in Section 12: the model is uncalibrated, and translating its variables into measurable proxies is a substantial empirical challenge. Nevertheless, the structural insights—that factors interact multiplicatively, that thresholds are modifiable, that the binding constraint matters most—provide a principled foundation for tool design that purely correlational approaches lack.

15 Conclusion

This paper has presented a dynamical-systems framework for modelling the realization of cognitive potential, defining the *genius regime* as a structurally stable region in a five-dimensional state space where plasticity, practice, environment, self-belief, and motivation jointly satisfy viability conditions.

The main results are:

(i) The genius regime requires the simultaneous satisfaction of five necessary viability conditions (V1–V5), formalizing the intuition that genius depends on the conjunction of many factors rather than any single trait (Proposition 4.3).

(ii) Self-belief acts as a local bifurcation parameter controlling the transition between a stagnation basin and a growth trajectory, with the critical threshold S_c depending on environment and motivation (Proposition 5.1).

(iii) Historical convergence of discoveries can be modelled as a threshold phenomenon on knowledge graphs, with co-discoveries becoming structurally inevitable above a critical knowledge density (Conjecture 7.2).

(iv) AI augmentation is limited by the bandwidth of the human–AI interface, with the integration coefficient approaching unity only as bandwidth increases (Proposition 8.3).

(v) Creativity is maximized at an interior optimum between divergent generation and convergent filtering, with the effective creativity entering the output growth equation as an amplification factor (Proposition 9.2).

We emphasize that this is a *conceptual model*—a mathematical language for organizing and connecting the rich empirical findings reviewed in Section 2—not a calibrated predictive theory. The model's value lies in three contributions: first, it provides a coherent formal structure in which the disparate factors studied by genetics, psychology, neuroscience, sociology, and AI research can be discussed within a single framework; second, it generates structural hypotheses (the conjunction requirement, the self-belief bifurcation, the bandwidth bottleneck) that can guide empirical research and intervention design; and third, it identifies the qualitative features—two attractors, multiplicative coupling, positive feedback—that are robust across a broad class of functional forms (Section 12).

The model also has clear limitations, acknowledged in Section 12: it is deterministic, treats the individual in isolation, does not specify parameter values, and has not been empirically calibrated. These limitations define the agenda for future work, which should include stochastic extensions, multi-agent models, empirical estimation of key parameters, and numerical viability analysis.

Within the Unified Structural Theory of Complex Systems, this paper fills the gap between the abstract dynamical core and the concrete applications, showing that the realization of intellectual potential is itself a viability problem amenable to the same structural analysis that has been applied to physical, biological, and social systems.

Perhaps the most constructive implication of the framework is that the bifurcation threshold S_c is not fixed by biology but is a function of controllable environmental and motivational variables. This structural observation—that the path to the genius regime depends on a *modifiable* threshold—is independent of the specific parameter values and survives under alternative functional forms. It suggests that interventions targeting environment, motivation, and self-belief can be designed on principled structural grounds, complementing the empirical traditions of educational psychology and expertise research.

In the language of the book that inspired this formalization [68]: geniality is not a gift of heaven, but a choice. The mathematics developed here does not prove this claim—no model can—but it shows that the claim is consistent with a formally coherent, structurally robust dynamical framework grounded in the relevant empirical literature.

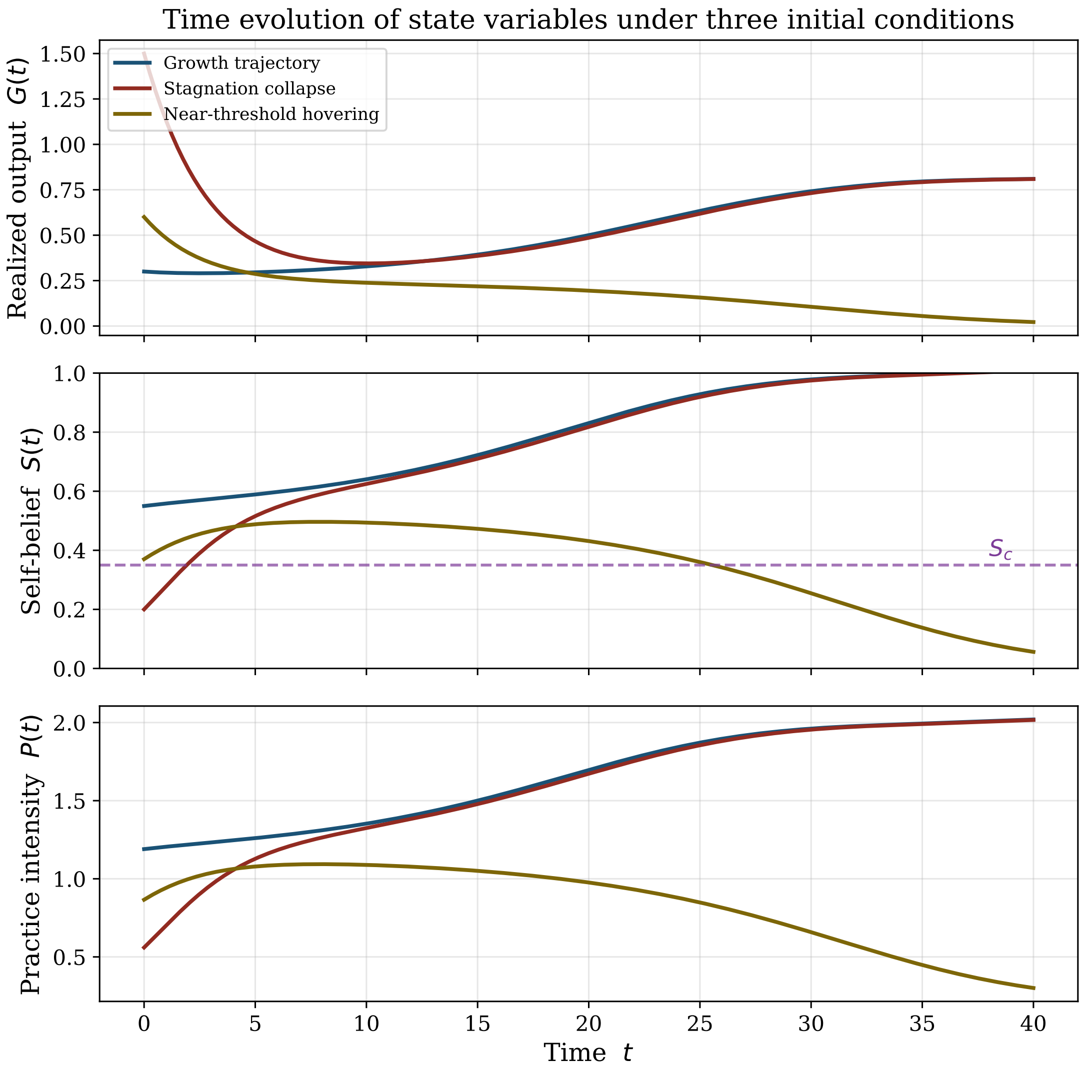

Figure 3: Time evolution of realized output $G(t)$, self-belief $S(t)$, and practice intensity $P(t)$ under three initial conditions. The growth trajectory (blue, $S(0) > S_c$) shows monotonic increase toward the growth attractor. The stagnation trajectory (red, $S(0) < S_c$) collapses despite initially high G. The near-threshold trajectory (gold, $S(0) \approx S_c$) hovers before tipping into one basin. The dashed purple line in the middle panel marks S_c.

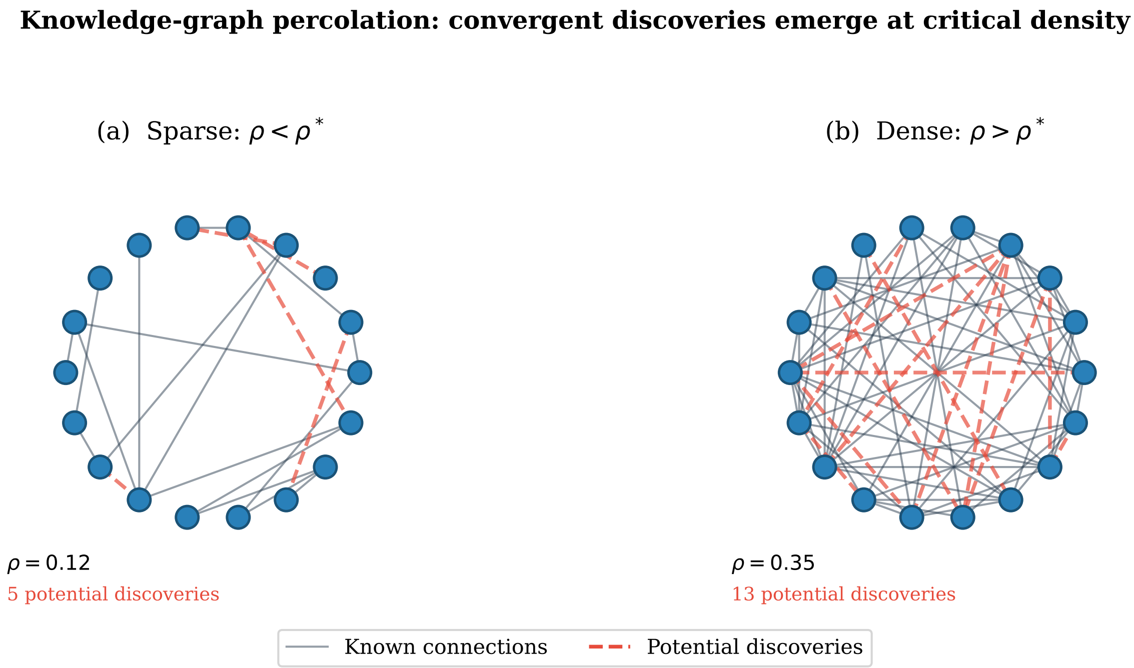

Figure 4: Knowledge-graph percolation illustration. (a) When the knowledge density ρ is below the critical threshold ρ^*, known connections (solid lines) are sparse and potential discoveries (dashed red lines) are isolated—simultaneous independent discovery is rare. (b) Above ρ^*, the dense network creates many short bridges between well-known concepts, and multiple potential discoveries become accessible to independent investigators simultaneously.

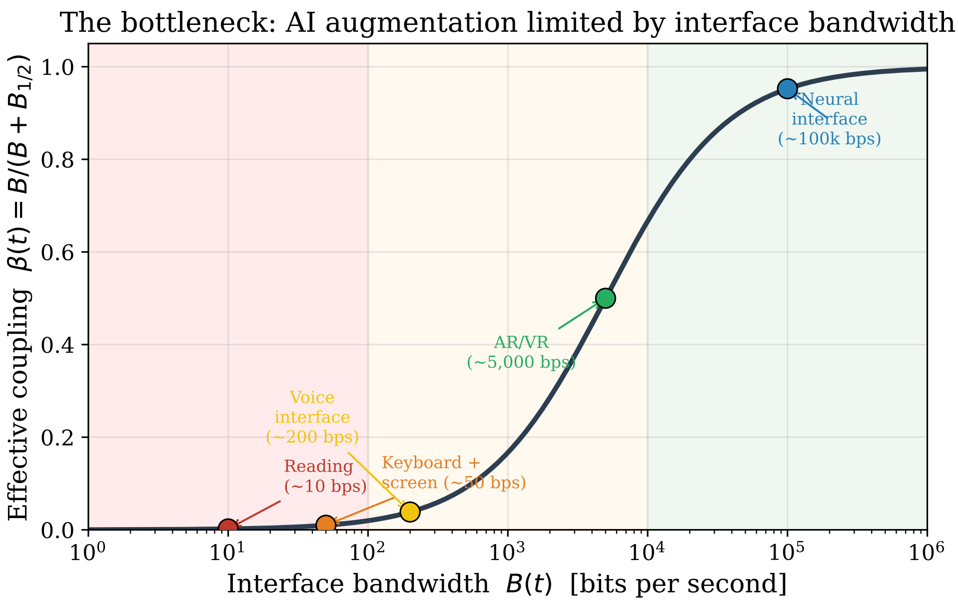

Figure 5: The bottleneck curve: effective coupling $\beta = B/(B+B_{1/2})$ as a function of interface bandwidth B (bits per second). Current keyboard-and-screen technology operates at approximately 50 bps, yielding $\beta \approx 0.01$—only a fraction of the AI's capacity is accessible to the human partner. As the interface improves (voice, AR/VR, and eventually neural interfaces), β approaches 1 and the AI's full computational power becomes available.

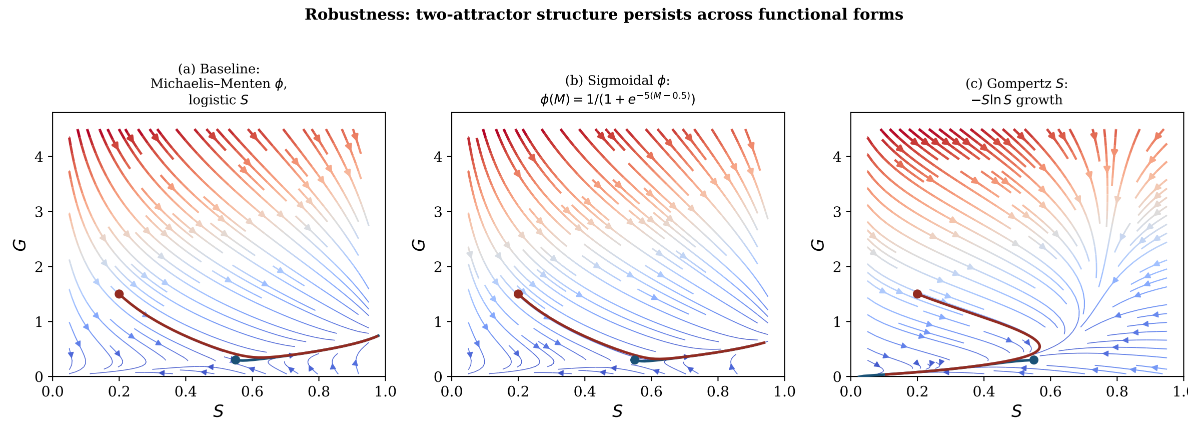

Figure 7: Structural robustness of the two-attractor dynamics under alternative functional forms. (a) Baseline model (Michaelis–Menten motivation, logistic self-belief). (b) Sigmoidal motivation function $\phi(M) = 1/(1 + e^{-5(M-0.5)})$, which introduces a sharper activation threshold. (c) Gompertz self-belief dynamics ($-S \ln S$ growth term). In all three panels, the two-attractor structure persists: a growth trajectory (blue) and a stagnation collapse (red) remain qualitatively similar despite quantitative differences in trajectory shape and attractor location.

Appendix A: Location within the Unified Structural Theory

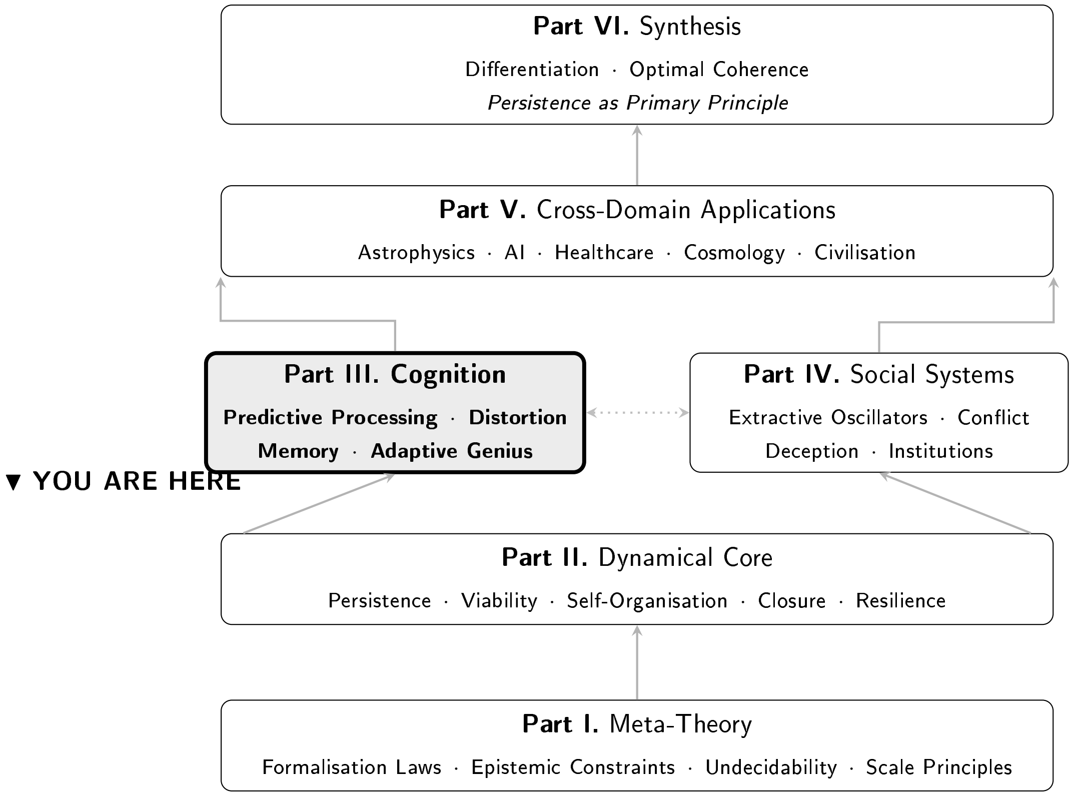

Figure 8: Location of the present paper within the Unified Structural Theory of Complex Systems (Kriger, 1999–2026). This paper develops the *Adaptive Genius* model within Part III (Cognition), drawing on the viability and persistence laws from Part II and connecting to extractive oscillator frameworks in Part IV and AI applications in Part V.

References

[1] de Groot, A. D. (1965). *Thought and Choice in Chess.* Mouton, The Hague.

[2] Chase, W. G. and Simon, H. A. (1973). Perception in chess. *Cognitive Psychology*, 4(1):55–81.

[3] Ericsson, K. A., Krampe, R. T., and Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100(3):363–406.

[4] Macnamara, B. N., Hambrick, D. Z., and Oswald, F. L. (2014). Deliberate practice and performance in music, games, sports, education, and professions: A meta-analysis. *Psychological Science*, 25(8):1608–1618.

[5] Hambrick, D. Z., Oswald, F. L., Altmann, E. M., Meinz, E. J., Gobet, F., and Campitelli, G. (2014). Deliberate practice: Is that all it takes to become an expert? *Intelligence*, 45:34–45.

[6] Plomin, R. and Deary, I. J. (2015). Genetics and intelligence differences: Five special findings. *Molecular Psychiatry*, 20(1):98–108.

[7] Bouchard, T. J. and McGue, M. (1981). Familial studies of intelligence: A review. *Science*, 212(4498):1055–1059.

[8] Okbay, A., Wu, Y., Wang, N., et al. (2022). Polygenic prediction of educational attainment within and between families from genome-wide association analyses in 3 million individuals. *Nature Genetics*, 54:437–449.

[9] Savage, J. E., Jansen, P. R., Stringer, S., et al. (2018). Genome-wide association meta-analysis in 269,867 individuals identifies new genetic and functional links to intelligence. *Nature Genetics*, 50(7):912–919.

[10] Simonton, D. K. (1999). *Origins of Genius: Darwinian Perspectives on Creativity.* Oxford University Press.

[11] Simonton, D. K. (2018). *The Genius Checklist: Nine Paradoxical Tips on How You Can Become a Creative Genius.* MIT Press.

[12] Eysenck, H. J. (1995). *Genius: The Natural History of Creativity.* Cambridge University Press.

[13] Hebb, D. O. (1949). *The Organization of Behavior.* Wiley, New York.

[14] Bliss, T. V. P. and Lømo, T. (1973). Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit following stimulation of the perforant path. *The Journal of Physiology*, 232(2):331–356.

[15] Doidge, N. (2007). *The Brain That Changes Itself.* Viking Penguin.

[16] Pascual-Leone, A., Amedi, A., Fregni, F., and Merabet, L. B. (2005). The plastic human brain cortex. *Annual Review of Neuroscience*, 28:377–401.

[17] Draganski, B., Gaser, C., Busch, V., Schuierer, G., Bogdahn, U., and May, A. (2004). Neuroplasticity: Changes in grey matter induced by training. *Nature*, 427(6972):311–312.

[18] Schultz, W., Dayan, P., and Montague, P. R. (1997). A neural substrate of prediction and reward. *Science*, 275(5306):1593–1599.

[19] Sutton, R. S. and Barto, A. G. (2018). *Reinforcement Learning: An Introduction.* MIT Press, 2nd edition.

[20] Raichle, M. E., MacLeod, A. M., Snyder, A. Z., Powers, W. J., Gusnard, D. A., and Shulman, G. L. (2001). A default mode of brain function. *Proceedings of the National Academy of Sciences*, 98(2):676–682.

[21] Buckner, R. L., Andrews-Hanna, J. R., and Schacter, D. L. (2008). The brain's default network: Anatomy, function, and relevance to disease. *Annals of the New York Academy of Sciences*, 1124(1):1–38.

[22] Beaty, R. E., Kenett, Y. N., Christensen, A. P., Rosenberg, M. D., Benedek, M., Chen, Q., Fink, A., Qiu, J., Kwapil, T. R., Kane, M. J., and Silvia, P. J. (2018). Robust prediction of individual creative ability from brain functional connectivity. *Proceedings of the National Academy of Sciences*, 115(5):1087–1092.

[23] Benedek, M., Jauk, E., Fink, A., Koschutnig, K., Reishofer, G., Ebner, F., and Neubauer, A. C. (2014). To create or to recall? *Neuroscience & Biobehavioral Reviews*, 46:311–322.

[24] Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review*, 84(2):191–215.

[25] Bandura, A. (1997). *Self-Efficacy: The Exercise of Control.* W. H. Freeman, New York.

[26] Dweck, C. S. (2006). *Mindset: The New Psychology of Success.* Random House.

[27] Stajkovic, A. D. and Luthans, F. (1998). Self-efficacy and work-related performance: A meta-analysis. *Psychological Bulletin*, 124(2):240–261.

[28] Yeager, D. S., Hanselman, P., Walton, G. M., et al. (2019). A national experiment reveals where a growth mindset improves achievement. *Nature*, 573:364–369.

[29] Sisk, V. F., Burgoyne, A. P., Sun, J., Butler, J. L., and Macnamara, B. N. (2018). To what extent and under which circumstances are growth mind-sets important to academic achievement? Two meta-analyses. *Psychological Science*, 29(4):549–571.

[30] Bravata, D. M., Watts, S. A., Keefer, A. L., et al. (2020). Prevalence, predictors, and treatment of imposter syndrome: A systematic review. *Journal of General Internal Medicine*, 35(4):1252–1275.

[31] Clance, P. R. and Imes, S. A. (1978). The imposter phenomenon in high achieving women: Dynamics and therapeutic intervention. *Psychotherapy: Theory, Research & Practice*, 15(3):241–247.

[32] Deci, E. L. and Ryan, R. M. (1985). *Intrinsic Motivation and Self-Determination in Human Behavior.* Plenum, New York.

[33] Ryan, R. M. and Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1):68–78.

[34] Csikszentmihalyi, M. (1990). *Flow: The Psychology of Optimal Experience.* Harper & Row.

[35] Csikszentmihalyi, M. (1996). *Creativity: Flow and the Psychology of Discovery and Invention.* Harper Perennial.

[36] Amabile, T. M. (1996). *Creativity in Context: Update to the Social Psychology of Creativity.* Westview Press.

[37] Guilford, J. P. (1950). Creativity. *American Psychologist*, 5(9):444–454.

[38] Guilford, J. P. (1967). *The Nature of Human Intelligence.* McGraw-Hill.

[39] Torrance, E. P. (1966). *The Torrance Tests of Creative Thinking—Norms—Technical Manual.* Personnel Press.

[40] Mednick, S. A. (1962). The associative basis of the creative process. *Psychological Review*, 69(3):220–232.

[41] Kenett, Y. N., Anaki, D., and Faust, M. (2014). Investigating the structure of semantic networks in low and high creative persons. *Frontiers in Human Neuroscience*, 8:407.

[42] Campbell, D. T. (1960). Blind variation and selective retention in creative thought as in other knowledge processes. *Psychological Review*, 67(6):380–400.

[43] Finke, R. A., Ward, T. B., and Smith, S. M. (1992). *Creative Cognition: Theory, Research, and Applications.* MIT Press.

[44] Martindale, C. (1999). Biological bases of creativity. In R. J. Sternberg (Ed.), *Handbook of Creativity*, pages 137–152. Cambridge University Press.

[45] Bronfenbrenner, U. (1979). *The Ecology of Human Development: Experiments by Nature and Design.* Harvard University Press.

[46] Heckman, J. J. (2006). Skill formation and the economics of investing in disadvantaged children. *Science*, 312(5782):1900–1902.

[47] Cunha, F. and Heckman, J. J. (2007). The technology of skill formation. *American Economic Review*, 97(2):31–47.

[48] Merton, R. K. (1968). The Matthew effect in science. *Science*, 159(3810):56–63.

[49] Merton, R. K. (1961). Singletons and multiples in scientific discovery. *Proceedings of the American Philosophical Society*, 105(5):470–486.

[50] Zuckerman, H. (1977). *Scientific Elite: Nobel Laureates in the United States.* Free Press.

[51] Ogburn, W. F. and Thomas, D. S. (1922). Are inventions inevitable? A note on social evolution. *Political Science Quarterly*, 37(1):83–98.

[52] Lamb, D. and Easton, S. M. (1984). *Multiple Discovery: The Pattern of Scientific Progress*. Avebury.

[53] Asch, S. E. (1951). Effects of group pressure upon the modification and distortion of judgments. In *Groups, Leadership and Men*, pages 222–236.

[54] Licklider, J. C. R. (1960). Man-computer symbiosis. *IRE Transactions on Human Factors in Electronics*, HFE-1(1):4–11.

[55] Engelbart, D. C. (1962). *Augmenting Human Intellect: A Conceptual Framework*. Stanford Research Institute.

[56] Card, S. K., Moran, T. P., and Newell, A. (1983). *The Psychology of Human-Computer Interaction*. Lawrence Erlbaum Associates.

[57] Carter, S. and Nielsen, M. (2017). Using artificial intelligence to augment human intelligence. *Distill*, 2(12):e9.

[58] Musk, E. and Neuralink (2019). An integrated brain-machine interface platform with thousands of channels. *Journal of Medical Internet Research*, 21(10):e16194.

[59] Clark, A. and Chalmers, D. (1998). The extended mind. *Analysis*, 58(1):7–19.

[60] Thelen, E. and Smith, L. B. (1994). *A Dynamic Systems Approach to the Development of Cognition and Action*. MIT Press.

[61] Port, R. F. and van Gelder, T. (Eds.) (1995). *Mind as Motion: Explorations in the Dynamics of Cognition*. MIT Press.

[62] Kelso, J. A. S. (1995). *Dynamic Patterns: The Self-Organization of Brain and Behavior*. MIT Press.

[63] Aubin, J.-P. (1991). *Viability Theory*. Birkhäuser, Boston.

[64] Aubin, J.-P., Bayen, A. M., and Saint-Pierre, P. (2009). *Viability Theory: New Directions*. Springer.

[65] Béné, C., Doyen, L., and Gabay, D. (2001). A viability analysis for a bio-economic model. *Ecological Economics*, 36(3):385–396.

[66] Martinet, V., Thébaud, O., and Doyen, L. (2007). Defining viable recovery paths toward sustainable fisheries. *Ecological Economics*, 64(2):411–422.

[67] De Cara, S., Houze, M., and Jayet, P.-A. (2009). Methane and nitrous oxide emissions from agriculture in the EU: A spatial assessment of sources and abatement costs. *Environmental and Resource Economics*, 32(4):551–583.

[68] Kriger, B. (2026). *How to Become a Genius? Neuroscience and Creativity*. Information Physics Institute.

[69] Kriger, B. (1999–2026). Unified Structural Theory of Complex Systems. 76 publications spanning seven domains. Full list at https://www.researchgate.net/profile/Boris-Kriger.

[70] Kriger, B. (2017). Four laws of self-organization. Zenodo. https://doi.org/10.5281/zenodo.18363729.

[71] Kriger, B. (2024). Transformational basis of persistence. Zenodo. https://doi.org/10.5281/zenodo.18435982.

[72] Kriger, B. (2026). Predictive processing derived from structural constraints. Zenodo. https://doi.org/10.5281/zenodo.18444910.

[73] Kriger, B. (2017). Extractive oscillators: A substrate-independent model. Zenodo. https://doi.org/10.5281/zenodo.18529185.

[74] Kriger, B. (2025). Binary-first star formation as persistence-selected outcome. Zenodo. https://doi.org/10.5281/zenodo.18155356.

[75] Kriger, B. (2024). Clinical discontinuity and AI: Restoring coherence in healthcare. Zenodo. https://doi.org/10.5281/zenodo.18395736.

[76] Kriger, B. (2024). No final theory: Scale-specific principles. Zenodo. https://doi.org/10.5281/zenodo.18099738.

[77] Kriger, B. (2025). Formalization laws. Zenodo. https://doi.org/10.5281/zenodo.18099527.

[78] Kriger, B. (2026). Definition-dependent provability. Zenodo. https://doi.org/10.5281/zenodo.18207348.

[79] Kriger, B. (2025). Law of imperative uncertainty. Zenodo. https://doi.org/10.5281/zenodo.18101601.

[80] Kriger, B. (2025). Law of limit to negation. Zenodo. https://doi.org/10.5281/zenodo.18101306.

[81] Kriger, B. (2026). Constraint–autonomy compatibility law. Zenodo. https://doi.org/10.5281/zenodo.18368991.

[82] Kriger, B. (2026). Viability mismatch law. Zenodo. https://doi.org/10.5281/zenodo.18433777.

[83] Kriger, B. (2026). Structural distortion principle. Zenodo. https://doi.org/10.5281/zenodo.18452700.

[84] Kriger, B. (2026). Representational isolation: Direct perception is formally impossible. Zenodo. https://doi.org/10.5281/zenodo.18331202.

[85] Kriger, B. (2025). Atemporal memory: Time is reconstructed at retrieval. Zenodo. https://doi.org/10.5281/zenodo.18381912.

[86] Kriger, B. (2026). Mental disintegration: DSM-5-TR mapped onto dynamical pathologies. Zenodo. https://doi.org/10.5281/zenodo.18556979.

[87] Kriger, B. (2005). Conflict as phase transition. Zenodo. https://doi.org/10.5281/zenodo.18361828.

[88] Kriger, B. (2026). Asymmetry of totalizing ideals. Zenodo. https://doi.org/10.5281/zenodo.18361829.

[89] Kriger, B. (2025). Deception framework: Perceived reality in multi-agent systems. Zenodo. https://doi.org/10.5281/zenodo.18526764.

[90] Kriger, B. (2020). Autonomy suppression in hierarchical systems. Zenodo. https://doi.org/10.5281/zenodo.18520653.

[91] Kriger, B. (2025). Dormant neutron star population constraints. Zenodo. https://doi.org/10.5281/zenodo.18409103.

[92] Kriger, B. (2026). Bayesian model comparison for cosmological hypotheses. Zenodo. https://doi.org/10.5281/zenodo.18230310.

[93] Kriger, B. (2026). Biospheric contribution to cosmic complexity. Zenodo. https://doi.org/10.5281/zenodo.18256772.

[94] Kriger, B. (2026). Local entropy inversion in AI systems. Zenodo. https://doi.org/10.5281/zenodo.18262199.

[95] Kriger, B. (2026). The inward turn: Computational efficiency as universal attractor (Fermi Paradox). Zenodo. https://doi.org/10.5281/zenodo.18437440.

[96] Kriger, B. (2026). The stimulus problem in post-scarcity environments. Zenodo. https://doi.org/10.5281/zenodo.18511908.

[97] Kriger, B. (2026). Differentiation as the ontological condition of actualization. Zenodo. https://doi.org/10.5281/zenodo.18268520.

[98] Kriger, B. (2026). Optimal coherence principle. Zenodo. https://doi.org/10.5281/zenodo.18341030.

[99] Kriger, B. (2026). Informational preconditions of meaning. Zenodo. https://doi.org/10.5281/zenodo.18292636.